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A Review of Nowcasting, Forecasting and AI in Economic Indicator Prediction

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Abstract

Accurate and timely assessments of economic indicators are essential to provide an early analysis of the economic situations. Therefore, forecasting and nowcasting applications with the Artificial Intelligence (AI) and Machine Learning (ML) technologies, which are offering enhanced accuracy and timeliness in predictions, have been widely used. The emerging chatbot technologies provide new insights in searching for data and generating information in these applications. This study conducts a comprehensive literature review to explore the use of AI and chatbot technologies in forecasting and nowcasting applications for economic and financial indicators. The "nowcasting", "forecasting", "artificial intelligence", "chatbot", "chat", and "bot" are searched on the Web of Science (WoS) database. The 111 studies on the topic are reviewed according to dependent variable, AI model, and main findings. The results indicated that the AI models increased the accuracy of the forecasting and nowcasting applications compared to the traditional time series models. However, the data availability remains a significant concern. The use of chatbots in predictions limited with the conceptual studies. Therefore, the forecasting performances of ChatGPT and Google Gemini in EU Gross Domestic Product (GDP- current prices) and unemployment rate were evaluated. They showed good predicting performances according to mean absolute percentage error (MAPE). The promising prediction accuracies provided by AI models can contribute to the decision making process of the policy makers, consumers, and investors. These findings can also be valuable guidance for the technical applications of AI and chatbot in forecasting and nowcasting.

Keywords Artificial Intelligence · Chatbot · Forecasting · Nowcasting

Extended author information available on the last page of the article

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1 Introduction

The methods and instruments used in applied economic research are rapidly evolving with new technologies (Leigh, 2024). The effects of these technological advancements are not limited to economic area, but also concern in education (Palasundram et al., 2020), health (Andrade-Arenas et al., 2024), agriculture (Ganeshkumar et al., 2023), and finance (Goodell et al., 2021).

Thanks to the improvements in the processing capacities of computers (or machines), the artificial intelligence (AI) based strategies has become a hot topic. The recent works indicate that the traditional predictors based strategies have decreased. However, the strategies based on the combination of AI and traditional predictors (in other words hybrid predictors) have risen (Nametala et al., 2023). The generative AI tools are one of the most intriguing of these advancements (Leigh, 2024). The Chat Generative Pre-Trained Transformer (ChatGPT), Gemini, and LLaMA are examples of the most used generative AI engines.

The ChatGPT 3.5 was developed by Open AI and released in 2002. It is an AI chatbot which has been trained by vast amount of textual data to generate human-like responses to input prompts (Singh & Singh, 2023). Google Bard (rebranded Google Gemini) is the competitor to the ChatGPT, with the learning capacity from different type of data such as text, video and audio (Ömür Arça et al., 2024). These large language models (LLM) are useful in code translating, data analysis, and mathematical model construction (Korinek, 2023). They provide 24/7 availability for free, quick responses to the prompts and easy to use (Ömür Arça et al., 2024). The accuracy, timeliness of information, ethic, and security issues related to using chatbots are open to discuss. However, the researches on chatbots have been rapidly increasing since 2021 (Lee et al., 2023). Therefore, it is worthwhile to conduct a review of the field.

In our study, we focus on the AI and chatbot based nowcasting and forecasting applications in economy. The accurate and timely assessments of economic indicators are essential to provide an early analysis of the current economic situations. Therefore, to review effects of these technologies on the economy will be useful for policy makers, public institutions, and academicians. The research questions which are focused on this study are:

> RQ1. What are the AI and chatbot based techniques used for forecasting and nowcasting applications in economy?
> RQ2. In which areas of the economy can AI and chatbot technologies improve forecasting and nowcasting applications?

For this purpose, the Web of Science (WoS) database is searched with the combination of "forecasting", "nowcasting", "chatbot", "artificial intelligence", "bot", and "chat" keywords. The initial research is concluded with 399 papers. After the screening and reviewing abstracts of 399 papers, the 111 papers are constituted the sample of the study. The overview about the papers is presented according to publication years, publication titles, and the number of citations. Then, the papers are divided classes by which methodology to track, which dependent variable to use, and which model to apply. According to result of this classification procedure, the 'cryptocurrency',

'bankruptcy', 'stock market index', 'oil and electricity prices', 'exchange rates', 'Gross Domestic Product (GDP)', and 'unemployment rate' are identified as widely considered dependent variables in the literature. Therefore, the studies considered these financial and economic variables are summarized according to research fields. Our results show that, the most used AI methods are Artificial Neural Network (ANN), Long-Short Term Memory (LSTM), Gated Recurrent Unit (GRU) and Support Vector Machine (SVM) in forecasting applications. To provide clean data and optimize the configuration parameters in neural networks (NN), the hybrid models are constructed with particle swarm optimization (PSO), genetic algorithm (GA), and random forest (RF) most frequently.

The number of forecasting applications is more than nowcasting applications in our review pool. This situation stems from the timely data availability. While the literature related to AI methodologies is expended both conceptual and empirical framework, the literature related to chatbots is limited with the conceptual one. Therefore, the forecasting examples with ChatGPT 5.2 and Gemini AI Pro are presented in our study. The most popular two chatbots are selected for this empirical analysis. The European Union (EU) 'GDP (current prices)' and 'unemployment rate' are predicted. The results indicate good forecasting performances of chatbots.

This literature review presents valuable guidance about the technical applications of AI and chatbot technologies in forecasting macroeconomic indicators. However, these applications should be supported with the AI strategies of governments. Especially, the data privacy should be the main concern for these strategies. On one hand, the easiness in availability the data provides the high accuracies and low error metrics in forecasting and nowcasting applications. On the other hand, the openness and the free use of the data should not violate the protection of personal data. Increasing the literacy in technology and economy should also be supported by the AI strategies of governments. The forecasting and nowcasting results with chatbots can include bias. Therefore, the users of chatbot assistant should have the ability of interpreting the suggestions of these models especially for economic and financial decisions.

This study is constructed in four sections. The sample and methodology of the study are presented in Sect. 2. The results of the literature review are given in Sect. 3. The last section summarizes the findings of literature review. It also presents the limitations and contributions of the study.

2 Sample and Methodology of the Literature Review

The literature review was adopted to present a detailed summary of the technical applications of AI and chatbot technologies in forecasting macroeconomic indicator. The detailed explanation of the search strategy is given in this section. The steps of the literature review process are presented in Fig. 1.

2.1 Step 1: Article Identification through WoS Database Search

Since the literature available on internet is not authentic and reliable, the most effective and economic way for reviewing literature is using scientific databases (Ganeshkumar

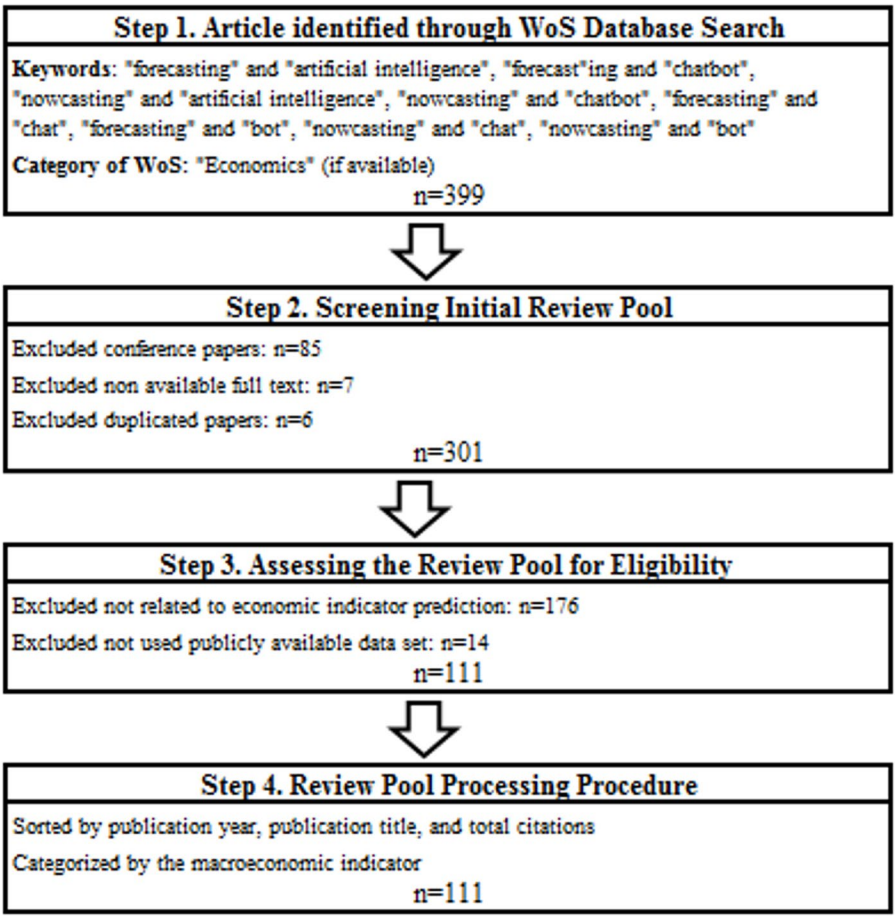


Fig. 1 The steps of the literature review methodology

et al., 2023). Therefore, the WoS database was used for literature review in this study. It has approximately 148 million documents such as journals, books, and proceedings dating back to the beginning of the twentieth century (Liu & Duffy, 2023).

The database was searched with the combinations of "forecasting" and "nowcasting" keywords with "artificial intelligence" and "chatbot". To prevent deformations, "chat" and "bot" keywords were also used. The WoS category was restricted with "Economics" for the ("forecasting" and "artificial intelligence") and ("forecasting" and "bot") search queries. For remaining search queries, there were not any research papers categorized in "Economics" by WoS database. Therefore, all searching results for these search queries were added to initial review pool.

2.2 Step 2: Screening Initial Review Pool

The selected keywords were searched by using "all fields" query in WoS. The initial search was resulted in 399 papers, for ("forecasting" and "artificial intelligence") 225

studies, for ("nowcasting" and "artificial intelligence") 112 studies, for ("forecasting" and "chatbot") 27 studies, for ("forecasting" and "chat") 28 studies, for ("nowcasting" and "chat") one study, for ("forecasting" and "bot") 4 studies and ("nowcasting" and "bot") 2 studies. In screening process, the conference papers (85 studies), papers whose full texts are not available (7 studies), and duplicated papers (6 studies) were excluded from the sample.

2.3 Step 3: Assessing the Review Pool for Eligibility

The abstracts of the remaining 301 studies were reviewed. The 176 of them included analyses not related to macroeconomic indicators. They proposed studies on weather forecasting, demand forecasting for tourism sector or specific products in fashion, sports- betting and movie audience. In addition, the 14 papers did not use publicly available data set. They used the data sets provided by the packages such as R and Kaggle. For these reasons, these 190 papers were excluded from the review pool. The remaining 111 papers constructed the data set of our review.

2.4 Step 4: Review Pool Processing Procedure

In the last step, the 111 papers were sorted by publication year. It provides to show the trend of the number of publications related to forecasting with AI and chatbot technologies in economics. Then, to analyze the influence of publications and journals in the topic, the publication sources and total citations were presented.

The full papers of these 111 research studies are reviewed and categorized in the type of analysis (empirical or conceptual) and dependent variable in the empirical analysis. The number of studies related to 'electricity and oil prices', 'stock market index', 'exchange rate', 'bankruptcy', 'cryptocurrency', 'GDP', and 'unemployment rate' are high in the review pool. It means that the forecasting and nowcasting applications in economy were focused on these research areas. Therefore, the objectives, methodology, and results of the studies in these research areas were presented under subheadings. The methods and results of the remaining articles were summarized in Table 5 (in Appendix). The number of papers including conceptual analysis of chatbot forecasting applications exceeded the number of empirical analyses. Therefore, the examples of GDP (current price) and unemployment rate forecasts were presented to evaluate the forecasting performance of ChatGPT 5.2 and Gemini AI Pro.

3 Results and Discussions

The statistical analyses of the review pool according to publication year, publication source, and total citations are given in this section. Then, the results of the review processes are presented for forecasting and nowcasting with AI models and chatbot technologies, respectively.

3.1 Statistical Analysis of Review Pool

To analyze the trend in the number of publications related to forecasting with AI and chatbot technologies in economics, Fig. 2 presented the number of publications for each year in our review pool.

There are three articles in our sample which are published before 2000s. All of these studies focused on financial indicators such as foreign exchange market (Beltrametti et al., 1997) and stock price index (Kim & Chun 1998;Kaboudan, 1999). Beltrametti et al. (1997) explained the reasons of selecting AI models as 'the inconclusive results of the rational-expectations models' and 'the need for cognitive view for forecasting and learning activities'. While Kim &Chun (1998) proposed new method to handle the bipolar output of the classical probabilistic NNs, Kaboudan (1999) proposed new measure for time series prediction with genetic programming (GP) approach. The findings of these studies were summarized in Table 5 (in Appendix).

Especially, after the 2019, the number of publications shows the increasing trend. It reaches its maximum with 33 papers in 2024. If this trend continues, the number of publications in 2025 will exceed the number of publications in 2024.

The influence of journals that discusses forecasting and nowcasting applications with AI tools in economics is also analyzed. The five journals consist of 40% of the articles in our review pool. In line with the selected keywords in our methodology, the top five journals come from forecasting and economy field. The number of articles according to these journals is presented in Fig. 3. The main academic journal is "Journal of Forecasting" in the review pool. The "Computational Economics" follows the main journal with 10 papers.

Finally, we concentrate on the most influential publications in our review pool. The list of 'most frequently cited 10 papers according to average per year scores' and 'the number of citations for the last five years' are given in Table 1.

The most frequently cited paper belongs to Petropoulos et al. (2022). They proposed a comprehensive nonsystematic review on forecasting. First, the theory of

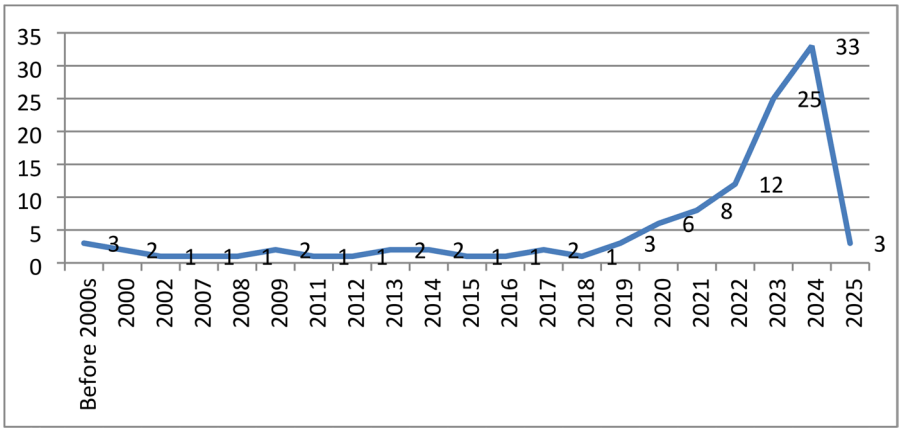


Fig. 2 Distribution of papers based on publication date

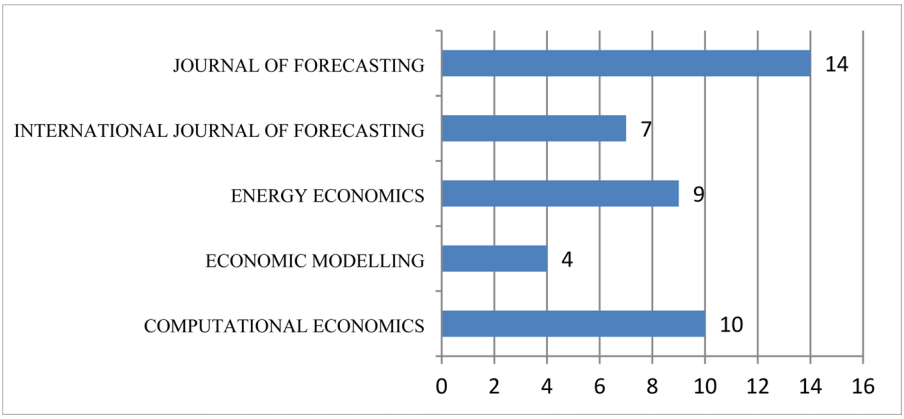


Fig. 3 Distribution of papers based on publication title

the forecasting was introduced. Then, methods for pre-processing data, statistical, econometric, combining, aggregation, and judgment models were explained. The methods for evaluating, validating and calibrating forecasts were given. At latest, the real-life applications were proposed under the research areas such as environmental, economic and financial applications. The encyclopedic nature of this study has enabled to become a reference for forecasting studies. It receives 284 citations in three years.

The 17 papers in our review pool included conceptual analysis or literature review. The nine of them reviewed the AI applications in agriculture (Leigh, 2024; Ganeshkumar et al., 2023), energy (Kuzmenko et al., 2024; Travieso-Gonzalez et al., 2024), finance (Goodell et al., 2021; Barboza et al., 2023), and international trade (Ozturk, 2024). The human bias in AI models was evaluated by Nguyen (2024). Grant & Üngör (2024) considered the AI revolution with 21 st century skills. Petropoulos et al. (2022) focused on the forecasting theory. The remaining seven studies include the chatbots related reviews such as factors affecting willingness to use chatbots (Lee et al., 2023; Camilleri, 2024; Baabdullah et al., 2022; Ullah et al., 2024; Thongsri et al., 2024), NLP in finance (Du et al., 2025), and comparison of chatbots performances in spatial tasks (Hochmair et al., 2024).

The 94 papers out of 111 papers in our review pool proposed empirical analysis. The empirical studies are categorized according to the research field. The forecasting and nowcasting applications with AI are conducted on 'stock market index (23 studies)', 'electricity and oil prices (11 studies)', 'cryptocurrency (6 studies)', 'exchange rates (6 studies)', 'bankruptcy (6 studies)', 'unemployment rate (4 studies)', and 'GDP (5 studies)' most frequently. The findings of selected studies which have most citations are presented under the subsections, specific to these research fields. The dependent variable, forecasting or nowcasting method, and the findings of the remaining empirical papers are given on the Table 5 in Appendix.

Table 1 The most frequently cited papers in our review pool

Authors	<=2021	2022	2023	2024	2025	Total Citations	Average per Year
Petropoulos at al. (2022)	0	46	112	122	4	284	71
Goodell et al. (2021)	7	55	84	108	3	257	51.4
Kraaijeveld & De Smedt (2020)	32	45	55	55	2	189	31.5
Yu et al. (2019)	72	28	25	12	0	137	19.57
Camilleri (2024)	0	0	0	27	2	29	14.5
Ding & Zhang (2023)	0	0	9	31	3	43	14.33
Baabdullah et al. (2022)	0	3	20	25	0	48	12
Li et al. (2019)	24	19	15	16	0	74	10.57
Yu et al. (2014)	67	10	9	11	0	97	8.08
Jursa & Rohrig (2008)	115	6	4	6	0	131	7.28

3.2 The Forecasting and Nowcasting Applications with AI in Our Review Pool

The theoretical information and the summary of the studies which included conceptual analysis on AI were presented in this section. In addition, the studies which included empirical analysis with AI models were summarized according to their research fields.

3.2.1 AI Related Studies: Conceptual Framework

The academic studies related to AI date back to the 1950 s with the idea of Alan Turing which proposed the machines being able to replicate human thinking processes (Ozturk, 2024). This technical innovation enables computer systems to accomplish activities that traditionally require human intelligence, such as learning, reasoning, problem solving, perception, and language understanding (Fernandez, 2019). Thanks to the advantages of 'easier access to data', 'increased data storage and computational processing capacities', and 'reduced costs of using this technology' (Fernandez, 2019), AI has gained popularity. It has been used in different sectors from agriculture to economy.

The Goodell et al. (2021) and Ganeshkumar et al. (2023) proposed the literature review on the AI for different sectors. Goodell et al. (2021) focused on the foundations, themes, and research clusters of AI and ML in finance. The Scopus database was used with search terms such as 'artificial intelligence', 'machine learning', 'data mining', and 'data science'. They reviewed 283 papers. The co-citation analysis revealed that AI and ML research in finance constructed nine foundational clusters, namely 'asset pricing and valuation', 'text mining and sentiment analysis', 'option pricing and valuation', 'financial fraud detection analytics', 'crowd funding', 'financial risk', 'credit risk', 'corporate failure and bankruptcy', 'portfolio optimization', 'forecasting and predictive analysis', and 'algorithmic'. The ANN, SVM, and ensemble models (EM) were widely used forecasting methods in finance.

Ganeshkumar et al. (2023) focused on AI applications in agricultural value chain. The 88 articles from WoS, Google Scholar, and Scopus databases were reviewed. They categorized the articles, like Goodell et al. (2021), according to agricultural stages. The results showed that AI related technologies were used for water quality, energy efficiency, optimization of fertilizer/pesticide usage and crop planning in 'farm inputs' stage. The drone, robot, and sensor technologies were generally used for cultivation and harvesting processes in 'farm production', 'quality tests', 'storage', and 'distribution' stages.

Barboza et al. (2023) also proposed the AI literature about the finance sector. Unlike the Goodell et al. (2021), they focused on the AI quality in forecasting asset prices. The 109 studies from 10 databases which are WoS, Scopus, Wiley, Emerald, Science Direct, Inderscience, Springer, SAGE, Taylor & Francis, and IEEE. The search terms of the study were 'finance', 'predict', 'forecast', 'machine learning', and 'artificial intelligence'. The most used methods are SVM, ANN, and auto regressive integrated moving average (ARIMA) as Goodell et al. (2021) indicated. However, the hybrid models which combine the artificial and non-artificial techniques showed increasing trend. The number of researches which have quarterly or annually

time frame were more than monthly or weekly time framed studies because of the data availability problems.

In addition to benefits of AI technology for different sectors in accomplishing different tasks, it has some limitations (Ozturk, 2024; Fernandez, 2019). AI technology has not still perfectly mimicked human cognitive capacities in complex thinking, profound knowledge, and common sense. It also needs big datasets to work efficiently, which raises concerns about data quality and bias (Ozturk, 2024). The anchoring is one of the potential sources of the bias in human decision making. Nguyen (2024) investigated the anchoring in S&P 500 index, federal funds rate, 10 year Treasury bond yield, EUR/USD and BTC/USD exchange rates by GPT-4, Claude 2, Gemini Pro, and GPT-3.5. In the first step, the 100 calibration forecasts were requested from these chatbots. In the experiment step, either 'a low anchor which corresponding to the 15th percentile of all calibration forecasts' or 'a high anchor which corresponding to the 85th percentile of all calibration forecasts' were given to the chatbots. Then, the forecasts are requested for each variable. The Gemini Pro's BTC forecasts, GPT-3.5's S&P 500 forecasts and GPT-3.5's BTC forecasts did not show statistically significant anchoring. For the models, which show statistically significant anchoring, the 'chain of though' and 'ignore previous' approaches were implemented for mitigating this effect. However, these approaches failed in minimize the anchoring effect especially in GPT-3.5 forecasts. Therefore, the users of AI assistant in financial decision making should consider the potential risk associated with the bias.

3.2.2 AI Related Studies: Empirical Framework

The studies are classified according to their research area. The research areas correspond to the subsections. This methodology will provide to analyze the AI applications in forecasting and nowcasting macroeconomic indicators in deeply. The forecasting and nowcasting methods which are frequently used within specific research areas can be analyzed. The similarity or differences between the findings of studies can be considered.

Stock Market Field The effect of unprecedented conditions such as pandemic, natural disasters, and wars cause the unprecedented changes in economy. Therefore, the predicting economic and financial time series is a difficult task (Al-Khasawneh et al., 2024). One of the challenging financial time series for prediction is stock market index. In addition, 'the difficulties related to choose appropriate forecasting or nowcasting methodology', 'choosing parameters of the model' and 'data availability statements' affect the accuracy of the predictions (Al-Khasawneh et al., 2024). The AI models have advantages for handling these challenging issues. They are preferred widely in forecasting stock market indexes, stock prices and stock returns (Metlek, 2022).

Aznarte et al. (2011) presented fuzzy rule based model for linguistic time series on IBM Down Jones Industrial Average Index. The performance of the proposed model was compared with the AR, smooth transition autoregressive model (STAR), and neuro coefficient smooth transition (NCSTAR) model. The all nonlinear models

outperformed the AR linear model. This result indicated that capturing the difference in stock index was difficult with linear methods. The performances of the STAR and NCSTAR were nearly similar according to RMSE, MAE and normalized MSE metrics both in sample and out sample forecasts. The proposed method provided nearly 19% improvements in the MAE in testing sample against the STAR model.

The type of the time series in Gonzalez & Arroyo (2012) is different from Aznarte et al. (2011). Gonzalez & Arroyo (2012) modeled histogram time series and interval time series with smoothing (in linear dependence) and k- nearest neighbour (K-NN) (in both linear and nonlinear dependence) methods. The autocorrelation in 5-min returns to S&P500 index for 2007 and 2008 were investigated. For each day in this period, the daily histogram was constructed. The high/low activity clusters produced a strong, positive and persistent autocorrelation for histogram time series. For interval time series, [90,100%] interquartile showed very strong autocorrelation. Although the exponential smoothing and K-NN methods did not show good performances, the better autocorrelation capturing performances of these methods were promising.

Sun et al. (2021) emphasized the structural changes impact on the time series forecasting. The existing model averaging approaches have designed with constant combination weights. The proposed local constant model averaging estimator selects optimal time-varying combination weights. The S&P500 price index was used as a dependent variable while 14 financial and economic variables were selected as inputs. In most cases, the proposed estimator was the best estimator and followed by jackknife model averaging (JMA) estimator. The 'Mallow model averaging (MMA)', 'smoothed Akaike information criterion (SAIC)', 'smoothed Bayesian information criterion (SBIC)', 'nonparametric version of bias-corrected AIC model', and 'nonparametric version of smoothed AICc (SAICc)' model averaging estimators were also considered. Among these estimators, the AICs estimators showed the worst performance.

Ballesteros & Miranda (2024) and Park et al. (2024) also focused on S&P500 price index. However, the selected input variables are in different nature. The input variables both in quantitative nature and qualitative nature were used in multilayer perceptron ANN model in Balleteros & Miranda (2024). The daily closing, opening, high, low, volume, historical reports of each company (financial summary, balance sheet, cash flow and income statement) were constructed the variables with quantitative nature. Total positive sentiment and negative sentiment opinions (social media sentiments) were the qualitative input variables. The most reliable model for technology, industrials, communication services, and consumer discretionary sectors was included the technical analysis indicators (with R^2 changing from 83.255% to 89.051%). The technical analysis and market sentiment indicators showed better performance in predicting stock prices in healthcare (with R^2 88.4185) and consumer staples sector (with R^2 87.500). These findings indicated that using qualitative nature inputs in predicting stock index give closer results to future performances of stock index.

The input variables of the model in Park et al. (2024) are stock chart images. The unknown dynamics in S&P500 index European call options for pre-Covid19 and post-Covid19 periods were modelled. The stock prices charts were converted into the image representations. These representations were used to train with deep generative

networks, which include the autoencoders, generative adversarial networks and diffusion models. When the number of images in the sample increases from 100 to 5000, the expected mean is closed to the true mean with low variance. The predictive capabilities of this sample size for different maturity levels were simulated. The proposed model with 20-day maturity show better performances than MLP, RNN, LSTM and ResNet with 0.41 MAPE and 0.8 MAE for pre-Covid19 period and 0.23 MAPE and 18.18 MAE for post-Covid19 period.

Thanks to the adherence to Islamic principles, being more stable and robust, Sukuk was received great attention by both the institutions and retail investors. The Sukuk index prices are evaluated by Katterbauer et al. (2022) and Metlek (2022). The Sukuk prices in China were estimated with deep learning method in Katterbauer et al. (2022). The data in quantitative nature and textual format were used for prediction with XGBoost method. The performance of the method was measured for 30 major Chinese companies in different payment schedules and forms. The RMSE values were mostly below 2% for all companies for training set while below 10% on the testing set. The result of the Armagan (2023a) supported the good performance of XGBoost method. The Borsa Istanbul IPO Index was evaluated in this study. While the MAPE value for ARIMA method was approximately 38.30%, it was 12.52% for XGBoost method.

Metlek (2022) predicted the price of Dow Jones Sukuk index with LTSM, RNN, and radial basis function neural network (RBFNN). The RNN models produced successful results in the forecasting of the time series. However, their performances could be affected by the lag length. To handle this limitation, a new RRN-based ANN model was also been developed. While the dependent variables were the Dow Jones "Middle East and North Africa" (DJMENA) and "Gulf Operation Council" (DJGCC) Sukuk indices, the independent variables were S&P Bonds, VIX, and DXY. The MAE values obtained from the DJMENA data set for LSTM, RNN, the proposed model, and RBFNN were 0.95, 0.32, 0.22, and 1.16 respectively. They were obtained for the DJGCC dataset as 1.24, 0.49, 0.20, and 1.58.

The study of Al-Khasawneh et al. (2024) attracts attention in the review pool because of focusing on the Pakistani stock market. The Pakistan stock exchange index (PSX-100) was predicted with deep learning methodology in this study. The LSTM-RNN was employed to increase the accuracy of the prediction. The effects of the convolution layer addition and the different optimizer usage were also analyzed. The traditional forecasting models from GARCH family showed moderate performance with RMSPE ranging from 0.269 to 0.412. The deep learning methodologies which used the LSTM- based architectures provides improvements in forecasting accuracy according to RMSPE ranging from 0.248 to 0.221.

Thanks to the data availability, the number of studies considered for the stock market area is the higher than the other research areas. The most of the studies were conducted on Down Jones, S&P500 and Borsa Istanbul stock indices. The hybrid models, fuzzy logic approaches, and simulations are most preferred models in forecasting stock prices. Especially, the NN models give the promised results both in accuracy and error metrics. The use of sentiment indexes and image data in forecasting models offers different perspectives for this research field.

Electricity and Oil Prices Field The electricity has certain characteristics in respect to no storage, short term and inelastic demand. Therefore, electricity prices are more volatile, randomness, intermittence and difficult to forecast (Jiang et al., 2023). In addition, 'the increasing penetration of renewable energy sources' and 'the liberalization of electricity markets' also cause the volatility and uncertainty in electricity prices. Therefore, predicting electricity prices accurately has become the concern for power producers, consumers, and traders in the energy sector (Agakishiev et al., 2025).

The study of the Kraft et al. (2020) has a broader scope which focusing on continental Europe. They tried to find best performing forecast method in terms of hyper parameters optimization and training strategies for modeling frequency containment reserve (FCR) prices. FCR is the first quality balancing reserve capacity with its shortest activation time. The prices of the largest European FCR market were forecasted with using autoregressive moving average (ARMA) models with exogenous predictors (SARIMAX). Also, the NN based models were used to compare the performance of proposed model. For NN models, the simple network structures outperformed the complex structures with 1.97%, 2.78%, and 2.89% for MAPE. These models also outperformed the SARIMAX whose RMSE was 136.82 and MAPE was 5.18%. However, SARIMAX method showed better performance than NN with rolling window training strategy and training hyper parameters fitting and over fitting.

Jiang et al. (2023) proposed new multivariable dual–output electricity price forecasting system to handle the inadequacies of existing single variable and single output models. The proposed system had three stages which were data pre-processing, combination forecasting and performance evaluation. In the second stage, BP, bidirectional LSTM and GRU models were used. The results of these forecasting models were combined with multi objective swarm algorithm. To evaluate the proposed multi input- multi output model performance, The New South Wales electricity market data (in Australia) was used. The MAPE of the proposed method was 1%–4% and the coverage rate of the interval forecasting exceeded 90%. Therefore, the designed system had superior ability in forecasting electricity price on account of multivariable load and price data arrangement rolling forecast mechanism.

Antoniadis et al. (2024) developed two methods based on hierarchical transfer learning for electricity load forecasting at UK. Then, the performance of these methods tested on French data during the Covid-19 lockdown and post-lockdown process. Thanks to the transfer learning process, the proposed generalized additive model (GAM) and RF model lead to improvements in the forecast in wider scale. With the multi scale information transfer through of experts, the quality of the forecasting increased. The MAPE and RMSE statistics indicated that the RF model was more efficient than GAM in forecasting UK data set. For French data set, GAM-RF improved the GAM performance with 50% and 45% decreases in MAPE for during and after the lockdown.

Agakishiev et al. (2025) also considered UK market in their study. The distributional multilayer perceptron (DMLP) model was proposed to predict the price data of the Nord Pool auction and the EPEX auction. In comparing the predictions, the realized quantiles histograms are used instead of the point estimates. While the DMLP approach underestimates the right tail highly according to Continuous

Ranked Probability Scores (CRPS) 16.28 EPEX and 19.80 Nord Pool, the LASSO QR regression provided a better fit for the tails. Despite the worse performance of NNs, they have smaller computational cost than LASSO QR approach.

In the oil price forecasting efforts, the hybrid method usages came to the front. Yu et al. (2014) proposed compressed sensing based denoising (CSD)- AI model to predict the crude oil prices. The CSD provided the clean data to AI tool with eliminating noises in data set. The proposed hybrid model significantly outperformed all other single models without CSD process and hybrid models with other denoising techniques, in terms of level and directional accuracies. Furthermore, in five-step-ahead prediction, CSD-ANN again achieved the highest Dstat and lowest RMSE values compared with the single and hybrid models. The novel CSD-ANN model showed best performances in the case of different data samples with different time ranges.

The findings of Li et al. (2019) supported the best forecasting performances of hybrid models for crude oil spot price. Two hybrid models, 'SVM optimized by genetic algorithm (GASVM)' and 'back propagation neural network (BPNN) optimized by genetic algorithm (GABP)', were proposed in this study. The hybrid models outperformed the single models (including GASVM, GABP and ARIMA models) in prediction accuracy. The VMD based hybrid models (including VMD-GASVM, VMD-GABP and VMD-ARIMA models) also outperformed the EEMD based hybrid models (including EEMD-GASVM, EEMD-GABP and EEMD-ARIMA models) in terms of MAPE and RMSE.

The oil and gas based predictors are indicated as most useful proxies for forecasting crude oil spot prices in Yu et al. (2014) and Li et al. (2019). However, the Google trends data was also contributed to the models. In the second study of Yu et al. (2019), oil consumption related Google Trend terms were used as independent variables. In the first step of the model, the granger causality test was applied to selecting Google Trend terms. In the second step, statistical and AI models were used to predict the global oil consumption trends and values. The classification models, which were LogR, BPNN, DT and SVM, and the forecasting models, which were LR, ELM, BPNN and SVR, showed better performances with Google Trends data according to PCC (AUC) values. In the case of oil consumption trend prediction, the PCC (AUC) values increased approximately 8.46% (0.00%), 1.17% (2.28%), 4.24% (16.22%) and 1.37% (21.66%) for LogR, BPNN, DT and SVM models respectively. When the forecasting performances of the AI techniques were evaluated, the SVR performed the best in terms of both MAPE and RMSE.

The studies in this research field indicated that AI models perform well in energy price prediction thanks to their vast amount of real time energy consumption data processing capabilities. This finding is line with the conceptual study of Kuzmenko et al. (2024). The NNs, decision trees, GA, and SVM were generally used methods in energy sector. The high accuracies of the hybrid methods in forecasting also draw attentions. The weather conditions, socio-economic variables, and energy demand were widely used variables in this research field. In addition to that, the contribution of the Google Trend data to track the oil consumption prefers different perspectives for predicting models. Thanks to the balancing performance between electricity production and consumption of AI, the dependence to traditional energy sources and greenhouse gas emissions could be reduced.

Cryptocurrency Field Digital currency based on the fundamentals of cryptography is known as cryptocurrency (Ali et al., 2024). They are growing faster than any other digital asset on the global financial market. Since the values of cryptocurrencies are influenced by a number of factors, the cryptocurrency markets are extremely volatile and prone to significant price changes in a short period of time. Therefore, it is challenging to forecast the future worth of cryptocurrencies. The accuracies of the predicting models are important issue to assess and manage the risks associated with the cryptocurrency investments (Ali et al., 2024).

Kraaijeveld & De Smedth (2020) investigated the Twitter Sentiment Index prediction performance of prices of the Bitcoin, Ethereum, XRP, Bitcoin Cash, EOS, Litecoin, Cardano, Stellar, and TRON. The Valence Aware Dictionary and Sentiment Reasoner (VADER) algorithm was used for calculating sentiment polarity scores. The 24,035,075 public tweets were classified as positive, neutral, and negative which are corresponding to the buy, hold, and sell recommendations. The calculated sentiment index was found as a granger cause of the returns of Bitcoin, Bitcoin Cash and Litecoin. The independent variables such as sentiment index, bulishness, and message volumes were more frequently causing variables than the dependent variables such as price returns and trading volumes of the cryptocurrencies. Therefore, it was concluded that Twitter Sentiment Index was slightly more of a "cause" than an ''effect" of the cryptocurrency market.

Since Li et al. (2023) only focused on the factors related to Bitcoin price; the study is not broader in scope as Kraaijeveld & De Smedth (2020). However, the financial market indicators and exchange rates were employed in the model besides the social sentiment index. The experimental results of the XGBoost and Shapley Additive exPlanation (SHAP) combination method showed that the most significant factors were S&P 500, WTI crude oil price and the USD/CNY exchange rate in Bitcoin price forecasting. Unlike the results of Kraaijeveld & De Smedth (2020), the social sentiment index was not existed among the top ten factors affecting Bitcoin price in Li et al. (2023).

Ali et al. (2024) used gradient boosting regression (GBR), RF and LSTM networks to understand the factors driving the performance of AI tokens. The prices of AGIX-singularity (NET), cortex (CTXC) and NMR were predicted with the daily closing prices of Bitcoin and Etherium, the Google and Amazon stock prices, crude oil and gold prices. The Defi tokens (Defi-BAT, Defi-MRK, and DefiSNX) and NFTs were the most important features which affect the AI based tokens prices. Among the proposed models, the LSTM model had more accurate and successful future predictions. This finding is not line with the Demirci & Karaatli (2023). They also used LSTM for predicting the price of Bitcoin, Etherium and Ripple. The LSTM model showed better performance only in predicting Etherium prices. For predicting Bitcoin and Ripple, the GRU model showed better performance than LSTM.

Behera et al. (2023) proposed a metaheuristic approach for forecasting cryptocurrency market data. The role of the metaheuristic algorithms which are fireworks algorithm (FWA), chemical reaction optimization (CRO), and the teaching–learning based optimization (TLBO), investigated the near optimal parameters related to hidden layer. The behavior of Bitcoin, Litecoin, Ethereum, and Ripple were

predicted with the proposed three hybrid models (ANN+FWA, ANN+CRO, and ANN+TLBO). The average performance measures of the ANN+CRO were 0.047 for MAPE, 0.038 for average relative variance (ARV), 0.039 for U-of-Theil statistic (UT), and 0.042 for normalized mean squared (NMSE). According to these statistics, ANN+CRO outperformed the other hybrid models. However, all hybrid models showed better performances than the GA and PSO models.

The focus of the Cohen (2023) is different from the other studies in this research area. The best cryptocurriency configuration which maximizes net profits was studied. For this purpose, the PSO method with average true range (ATR) concept was used. The daily price information of Bitcoin, Ethereum, XRP (Ripple), Binance Coin and Cardano were dependent variables in the analysis. The stand-alone optimized ATR systems showed better performance for long trades than short trades for each cryptocurriency. While the best configuration for Ethereum was 14 days which result in $5969 net profits, it was 12 days with $669 net profits for Binance Coin. If the optimized Keltner Channel based system was added to the proposed approach, the forecasting ability of the stand-alone optimized ATR systems were increased in terms of net profits and profit factors.

The papers in this research area have different purposes such as 'determining the factors effecting cryptocurrency prices', 'forecasting the cryptocurrency prices', and 'determining the best cryptocurrency configuration'. The most of them considered the Bitcoin and Etherium as cryptocurrencies. The methods used in the studies were varied from granger causalities to metaheuristic approaches. The hybrid models were also used in the analysis and showed better predicting performances than single models.

Exchange Rates Field Forecasting currency exchange rate has drawn a lot of interest due to its inherent complexity and real-world applicability (Nag & Mitra, 2002). The non-stationary natures of the exchange rates cause the linear unpredictability. To capture the non-linearity in the exchange rate series, random walk models and NNs are widely used. If AI models with straightforward train-validate-forecast methodologies are selected to predict exchange rates, the nonlinearity in exchange rates cannot be captured (Reikard, 2023). Therefore, self-adaptive models which enable the evolving of the weights or regression coefficients can provide more accurate predictions.

The hybrid AI methodology to model daily Deutsche mark/US dollar, the Japanese yen/US dollar and the US dollar/British pound rates was proposed by Nag & Mitra (2002). The GA was used for the optimizing NN architecture parameters. The performance of the proposed model compared with the conditional heteroscedastic models (ARCH, GARCH, AGARCH, EGARCH and GARCH-M) in terms of the average absolute error (AAE), MAPE, MSE, maximum absolute error (MaxAE) and R^2. The out-of-sample AAEs for the best performing genetic neural network were $6.06 \times 10\text{-}3$, 0.63, $5.1 \times 10\text{-}3$ for the Deutsche mark/US dollar, the yen/US dollar, and US dollar/British pound rates, respectively. The MAPEs were around 0.33%, 0.49%, and 0.31%. The results indicated that the proposed method outperformed both the traditional non-linear time series techniques and fixed-geometry NN models.

Nastac et al. (2007) also focused on neuro adaptive model which provide the dynamic nature for input variables, output variables, and time delays. This model had remembering process to preserve the relevant aspects from the previous training process. The searching for the optimal architecture took a relatively long time. Then, the system was easily retained. Thanks to the performance of the retaining phase, the test error decreased in forecasting Euro/Romanian currency exchange rate.

Kozlovskyi & Poliakov (2024) also focused on the one exchange rate which is US dollar/Ukrainian hryvnia (USD/UAH). However, they evaluated the effect of the war and used 16 factors grouped into economic, political, and military factors. Since the proposed model was based on the neurolinguistic, the exchange rate related input variables were also in qualitative nature. The result of the analysis indicated that the US dollar/Ukrainian hryvnia (USD/UAH) exchange rate could decrease more than twice within one year. This negative trend continued with the effect of the war.

The content of the Amelot et al. (2021) and Reikard (2023) were comprehensive than Nastac et al. (2007) and Kozlovskyi & Poliakov (2024) in terms of considering different exchange rates and different models. Amelot et al. (2021) constructed time series models including ARIMA, GARCH, 'ANN models including non-autoregressive models with exogenous input (NARX) neural network', and 'statistical topologies including the kernel principal component analysis (KPCA) using SVR' to forecast the EUR/MUR, GBP/MUR, CAD/MUR and AUD/MUR exchange rates in Mauritian forex market. The GARCH model showed best performance in terms of volatility clustering and prediction. However, the NARX and KPCA–SVR topologies outperformed the linear time series models. The NARX model also outperformed the KPCA–SVR model. The lowest MSE and RMSE statistics indicate that, the NARX models with LM and Bayesian regularization training algorithms were the most appropriate method of forecasting the different currency exchange rates.

Reikard (2023) compared the ANN, nonlinear regression, and RNN models performance to forecast exchange rates for Canadian dollar, British pound, Swiss franc, and Euro. For Canada dollar, the most accurate forecast was achieved at the 1-month horizon with 0.0146 MAE and 0.0193 RMSE. The perceptron model for the difference in 1 and 2 months were the most accurate forecasts for Swiss franc and Euro. However, for the difference in 3 months levels, the LSTM outperformed the perception model in forecasting Swiss franc.

The main focus of the studies is Euro currency generally in this research field. It is followed by the US dollar and British pound. The study of the Nag & Mitra (2002) concluded that the conditional heteroscedastic methods showed better performances than traditional non-linear time series and fixed geometry NN models. However, the performances of the fixed geometry NN models outperformed the linear time series and autoregressive models.

Bankruptcy Field A complex relationships network has emerged in the business world as a result of globalization (Korol, 2013). In addition, the global financial crisis, unprecedented situations such as pandemic and war have affected the financial standing of entities in market economy. Households' financial circumstances are also adversely affected by these situations because of rising interest rates, inflation, and unstable currency values (Korol & Fodadis 2022). Therefore, predicting personal and

market bankruptcy provide an effective early warning system to make credit decisions, preserve financial stability, and promote the general well-being of the economy (Letkovsky et al., 2023).

Hu & Ansell (2009) predicted US retail market default with sequential minimal optimization (SMO) method. The performance of SMO was compared with Naïve Bayes, logistic regression (LR), recursive partitioning and ANN according to classification accuracy rate and the area under the receiver operating characteristics (AUROC) curve. The 67 potential input variables, which are constructed two groups as internal and external factors, were considered with PCA for dimensionality reduction. The performances of the models varied according to different time periods and input variables. The overall classification accuracy rate among different credit scoring techniques was over 80.49% (AUROC was over 0.8456) in the year before financial distress. It was above 79.27% (AUROC was above 0.7345) for the 5 years before default.

Letkovsky et al. (2023) predicted bankruptcy in Slovak Engineering Industry with LR, ANN, and SVM methods. The sample included eight financial indicators from the indebtedness, liquidity, profitability, and activity categories for 825 businesses. The AI models showed better performances than LR model. While the average prediction accuracy is 93.86 for LR model, its capacity in discriminating bankrupt situations was 43.55% for training set and 44.44% for test set. The average prediction accuracy of the ANN was lower than the LR model as 85.12%. However, its capacity in discriminating bankruptcies was 74.07% for test set. The SVM showed the best performance in discriminating bankruptcy situations with almost 93%. Therefore, the SVM model was the most preferable model to predict bankruptcy.

Korol (2013) constructed early warning system for bankruptcy risk for European and Latin American enterprises. The six bankruptcy prediction models proposed which consists of discriminant analysis, decision trees, and ANN model for the analysis of one year and two years prior to bankruptcy. The decision tree model showed best performances for Latin American enterprises with 96.66% effectiveness for one year before and 95% for two year before. The discriminant analysis was less effective than the decision tree model both in one year before (86.53% effectiveness) and two year before (74.51% effectiveness). The effectiveness of decision tree model was also better than the ANN model. The findings of the Navarro-Galera et al. (2024) are line with the Korol (2013). The decision tree based model showed best performance in the early warning system for Spanish small and medium-sized enterprises (SMEs). The classification accuracy of the proposed model was above the 95% both one year and two year ahead predictions. The areas under the ROC curves were 94.2% and 92.7% for one year and two year predictions, respectively.

Korol & Fodadis (2022) focused on personal bankruptcies in Poland and Taiwan. They developed six models with using fuzzy sets and GA in the learning process of ANNs (GA-ANN). The 12 ratios, which are constructed with annual income, monthly income, the total value of assets held, the value of all loans taken, credit cards debt, the value of monthly interest rates paid, and demographic factors such as age, level of education, and number of children, were used as independent variables.

The models with fuzzy set learning process outperformed the other AI and statistical models in overall effectiveness and type-I error.

Brygala & Korol (2024) also studied on the personal bankruptcies. However, their analysis did not include the hybrid model like Korol & Fodadis (2022). The 957 bankrupts and 957 non-bankrupts data from the US Survey of Consumer Finances are the inputs of SVM, RF, AdaBoost, XGBoost, LightGBM, and CatBoost models. The LightGBM showed the highest total effectiveness (74.93%). While the lowest type-I error was also achieved by LightGBM (21.93%), the lowest type-II error was achieved by XGBoost (27.04%).

The classification methods for personal and market bankruptcies came to the forward in this research area for different countries such as US, Spain, Slovak and Taiwan. The accuracies of the SVM and decision tree models in the classification draw the attention. The number and nature of the input variables showed higher differences. Therefore, the dimensionality reduction models can be used to determine the important independent variables.

Unemployment Rate Field The necessity for more precise unemployment forecasting models has grown urgently. This situation stems from 'the COVID-19 pandemic's detrimental effects on the world economy', 'the rising jobless rates (Grybauskas et al., 2023)', and 'the technological advancements capabilities of replacing human labor (Mutascu & Hegerty, 2023)'.

The first empirical studies in our review pool for this research area is belong to Kreiner & Duca (2020). Two alternative methods which are PCA and lasso model were used for dimensionality reduction. To optimize the configuration of the ANN, the different type of activation functions, number of hidden layers, and number of neuron in the hidden layer alternatives were tested. The best configuration for a four-quarter ahead unemployment rate forecast used three hidden layers and 97 neurons per layer, with a size of 120 neutrons that used logistic activation function. This ANN minimized the RMSE at 0.20. For the lasso model, the penalty parameter was selected as 15. Then, the same ANN configuration was used for predicting unemployment rate and the RMSE was found as 0.31. The lasso model could also assess the importance of individual variables. For four-quarter ahead forecasts, the most informative variables were housing starts, the change in nonfinancial corporate commercial mortgages, and unfilled German job vacancies in order.

The content of the Mutascu & Hegerty (2023) was more comprehensive than Kreiner & Duca (2020). They were predicted the unemployment rates with ANN for 23 countries, over the period 1998–2016. A multilayer ANN perception model was considered with a feed forward back-propagation network type, "Levenberg–Marquardt" training function, "Gradient Descent" learning function and "Hyperbolic Tangent" transfer function. The volume of AI patents (as a proxy), inflation rate, population, labor productivity, government size, FDI, and the lagged unemployment rate were independent variables. The proposed ANN with one layer and 10 neurons showed good performance with an overall R^2 of 0.912. The AI input was a top contributor to the prediction of unemployment rate according to Garson's Algorithm.

The foreign direct investment, total population, labor productivity, and lagged unemployment rate had also big contribution to the prediction model.

The study of Nia et al. (2022) draws attention with using nowcasting model and considering the Covid19 pandemic. Twitter data based nowcasting application was used to predict the unemployment rate of South Africa during the Covid-19 pandemic. The unemployment-related tweets were used as inputs. The Principle Component Regression (PCR) performance in predicting unemployment rate were 0.92 for RMSE, 0.018 for MAPE, 0.018 for SMAPE, and 0.93 for R^2. While the volume of the tweets had positive correlation with the unemployment rate, the sentiments of the tweets showed a negative correlation.

The nowcasting applications of Grybauskas et al. (2023) also considered the effect of pandemic like Nia et al. (2022). However, they used Google Trend search queries as inputs for LSTM method. The 222 keywords were classified into six dimensions which are 'job search', 'mental health', 'violence and abuse', 'leisure search', 'consumption and lifestyle', 'disasters, war, and viruses'. With using different model construction parameters, the RMSE errors decreased to 16.7% for pandemic time frame, 25.23% for full timeframes, and 27.7% for pre-pandemic time frame. If the keywords performances were evaluated, the work-related keywords (such as unemployment and career builder) and the consumption lifestyle related searches (such as luxury yacht for sale and plastic surgery cost) were important variables. The mental health keywords such as 'anxiety' and 'withdrawal from friends' were also significantly improved the prediction accuracy of the model.

The content of the studies are comprehensive which includes different types of input variables and considers different countries in this research area. The nowcasting applications based on the Twitter data and Google Trend data also draw attention. While nowcasting errors ranges from 27.7 to 0.018 (RMSE), the forecasting errors ranges from 0.20 to 0.31 (RMSE) in studies. It means that the input data in qualitative nature (such as tweets and search queries) showed better performance in predicting unemployment rates.

Gross Domestic Product Field The GDP is often considered one of the important indicators to measure the economic development of the country (Sencan & Sencan Sahin, 2022). The forecasting and nowcasting GDP facilitate planning and decision making for firms, households, and policy makers.

The most of the studies in this research area focused on the effects of Covid19 pandemic on GDP in line with unemployment rate research area. While Rakha et al. (2021) assessed the economic impact of the Covid-19 in the UK with using time-series mining, Alfarra & Hagag (2024) predicted the influence of the Covid-19 on Arab economy with the ARIMA and LSTM model.

The GDP, GDP growth and the performance of the construction, production and service industries were analyzed in continuous and categorical time series in Rakha et al. (2021). For continuous time series analysis, the ARIMA model was selected while the Symbolic Aggregate approXimation (SAX) and Vector Space Model (VSM) combination model (SAX-VSM) was used for categorical time series analysis. The largest fall in quarterly GDP was −20.4% in the second quarter of 2020 which is

not seen over the last 60 years in the UK. According to the predictions, the decline in GDP was expected for 2021 Q1. For the outputs of the service and construction industries, it was observed that the decrease continued in 2021 Q2. However, GDP and the output of the manufacturing industry showed growth in 2021 Q2.

Alfarra & Hagag (2024) predicted the GDP growth rate for the next five years until 2025 based on the real data from 1968 to 2020. This study showed difference from the other studies in composition of the training and test set. The authors used the 90% of the data for training while the remaining 10% used for testing. The general trend in the studies in our sample for this ratio is 80:20. Since the LSTM based model was more fitted to the actual data with 0.0007 MAE and 0.02 MSE than ARIMA model, it was selected for the prediction. The result showed that the Covid-19 pandemic caused a decrease in GDP by approximately 17.22% and 5.41% in 2020 and 2021, respectively.

Cubadda et al. (2025) proposed the new model which is time-varying parameter factor augmented VAR with time-varying volatility (TVP-FAVAR-SV) to forecast US GDP, CPI, and federal funds rate (FFR). This method handled with the over parameterization in studying with big datasets. The initial estimation sample covered from 1960:Q1 to 1971:Q4. The forecast window covers the two oil shocks, Dot-com bubble, great recession, and Covid-19 pandemic. With three different variables, eight different forecast horizons, and three different forecast metrics, 21 models were constructed. While all models showed good performances in some cases, the TVP-FAVAR-SV models with 4 lags generally outperformed the competitors.

One of the limited nowcasting applications in our review pool is belong to Jönsson (2024). The nearest neighbour algorithm was implemented to nowcast Swedish GDP. The dimensions of the algorithm were the number of neighbours and distance metric. The Manhattan, L1 and L2 form distance metrics and two different initial sample periods were considered. The KNN method showed better nowcasting performance than AR and RW algorithms according to MAE and MSE indicators. However, the performance of the KNN algorithm was worse than RW model according to ME measure. The best performance was showed by the algorithm with Manhattan or L1 norm distance metric, equal weighting scheme, and 4 or 5 neighbours.

The results of the forecasting and nowcasting applications in GDP research area conducted for different countries such as UK, Türkiye, and Swedish. The study of the Alfarra & Hagag (2024) draw attention with the superior performance of LSTM. The effect of the Covid-19 is main concern of the most studies.

3.3 The Forecasting and Nowcasting Applications with Chatbot in Our Review Pool

The WoS database does not categorize any paper in “Economics” for (“forecasting” and “chatbot”), (“nowcasting” and “chatbot”), (“forecasting” and “chat”), (“nowcasting” and “chat”), and (“nowcasting” and “bot”) search queries. For, (“forecasting” and “bot”) search query, the “Economics” category is available. However, most of these papers are related to build-operate-transfer (bot) project. Therefore, the papers related to chatbots are very scant in our review pool. This limited literature is given following subtitles as studies focus on conceptual analysis and empirical analysis, respectively.

3.3.1 Chatbot Related Studies: Conceptual Framework

Chatbots are automated conversational systems that employ natural language processing (NLP) to communicate with human users (Liu & Duffy, 2023). They are initially intended to be the virtual assistant for the activities such as answering inquiries, providing driving directions, and customizing the user experience. Thanks to the advancements in AI, ML, and NLP, chatbots are now defined as advanced digital assistants. They perform a range of tasks, from simple command processing to serving as interactive agents (Liu & Duffy, 2023).

Since chatbots rely on the data which came from multiple sources such as Twitter, Wikipedia, and Amazon (Lin et al., 2023), the efficiency and capability of these technologies is limited to the quality of the data. The identifications of statements also effects to get correct and appropriate answers to the response (Lin et al., 2023).

The most popular chatbot application is ChatGPT (Lin et al., 2023) and is closely followed by Gemini, Copilot and Claude-3 (Hochmair et al., 2024). Leigh (2024) considered the reading data, interpreting data and drawing graphs performances of ChatGPT in agricultural productivity. The Australian Historical Agricultural Forecast Database was uploaded to the ChatGPT 4.0. Then, 'forecast horizon against the forecast error' and 'forecast error against the commodity' graphs were requested. Based on 14,622 observations in the data set, ChatGPT 4.0 showed that the forecast error decreased in forecasting horizon. The forecasting errors changed from 18% for mutton to 6% for poultry. The author requested some other graphs related to 'average errors for crops' and 'the forecast results which show greatest improvement' and 'the forecast results which show largest deterioration over time'. ChatGPT 4.0 also figured these graphs with a little mistake such as classifying American dollar as a dairy crop. However, it can be concluded that AI tool was useful for creating graphs and images. Also, it showed good performance understanding the features of the data set without training. Therefore, the author concluded that AI allows exploring the patterns in data without coding and has potential to democratize data analysis process.

The coverage of the Hochmair et al. (2024) is broader in scope than Leigh (2024). The comparison of the chatbots (ChatGPT-4, Gemini, Copilot, and Claude-3) correctness for 76 tasks across seven spatial and GIS-related categories was presented in this research. The seven task categories were 'spatial literacy', 'GIS concepts', 'mapping', 'function interpretation', 'code explanation', 'coding', and 'spatial reasoning'. While the spatial literacy category included 16 tasks, 10 tasks were included by each of remaining six categories. These 76 tasks included the 'open-ended questions', 'multiple choice questions', 'capabilities in generating map', 'explaining the purpose of functions', 'explaining the purpose of the program code', 'writing programming code', and 'reasoning'.

Since the Hochmair et al. (2024) provide good comparison about the capabilities of different chatbots, the findings of the study are given in detail. In total, ChatGPT-4 showed 76.3% correctness, followed by the Copilot (71.1%), Claude-3 (64.5%), and Gemini (55.3%). The ChatGPT-4 completed the tasks in GIS concept, function interpretation, and code explanation with 100% correctness. The Copilot showed 100% correctness for GIS concept and code explanation, Claude-3 completed all task correctly in only code explanation category. Gemini did not show 100% correctness for any

task categories. The lowest correctness were shown by ChatGPT-4 in spatial reasoning category (20%), Gemini in mapping category (0%), Claude-3 in mapping category (20%), and Copilot in mapping category (20%). If the correctness of the completed task were compared by categories, the highest percent was belonging to the GIS concepts and code explanation with 95%. The lowest percent correct was belonging to the mapping (25%) and spatial reasoning categories (32.5%). The relationships between the 'chatbot and response correctness' and 'task category and response correctness' were found statistically significant according to chi square testing. These significant differences could be originated from the architecture of chatbots (GPT architecture for ChatGPT-4, Claude-3, Co-pilot and Moe architecture for Gemini).

Baabdullah et al. (2022) questioned which factors shape the customers virtual flow experiences with AI chatbots. The flow experience was defined as the cognitively efficient, motivated and happy psychological state of the customer. The positive flow experience increased the interactivity of the customer in the digital platform. The survey sample included 500 Saudi customer of courier and package delivery. The effects of 'readability', 'transparency', 'personalization', 'ubiquitous connectivity', and 'responsiveness' on virtual flow experience were investigated by structural equation modeling (SEM). These five factors explained the 0.60 variance of the virtual flow. The strongest predictor of virtual flow was the readability. The effects of virtual flow experience on the communication quality and customer satisfaction were also evaluated and both links were found significant.

Camilleri (2024) investigated the impacts of 'performance expectancy', 'effort expectancy', 'perceived information quality', 'social influence', and 'perceived interactivity' on intentions to use ChatGPT. The effects of effort expectancy and source trustworthiness on performance expectancy were also evaluated. The sample was constructed by 654 staff and students in the Southern European University. The most significant link was found between trustworthiness and performance expectancy. The links between the 'perceived interactivity and intentions' and 'effort expectancy and performance expectancy' were also highly significant. The findings of Thongsri et al. (2024) are also line with the Camilleri (2024). The samples of these two studies are similar in terms of the number of participants and conducting at universities. The top three factors affecting the willingness to use ChatGPT were found as system quality, cognitive need, and information quality according to SEM and ANN models in Thongsri et al. (2024). Overall, the quality of information, communication and system are indicated the most influenced dimensions on ChatGPT usage by Baabdullah et al. (2022), Camilleri (2024), and Thongsri et al. (2024).

The co-citation and big data analysis on service chatbots were proposed by Lee et al. (2023). The chatbot-related 386 article gathered from WoS database and 4,840 newspaper articles gathering from Google news for this purpose. The most of the studies in review pool were related to 'communication', 'psychotherapy', and 'retailing' from the technology adaptation perspective. However, the literature was scant for environment, sustainability, law, and policy. The topic modeling results indicated that, the major development fields for service chatbot technologies were healthcare, technology and university. The problems or obstacles about the chatbots were privacy and the intended or unintended ethical problems caused by developers.

Ullah et al. (2024) and Nametala et al. (2023) focused on the chatbot usage in investment decisions. Ullah et al. (2024) investigated the ChatGPT usage dimensions and investment decisions of investors from the view of financial literacy. The sample included 388 fully completed questionnaires from the investors in Pakistan stock market. The results indicated that the 'analyzing data' and 'managing risk dimensions' had significant effects on investment decisions. The 'optimizing portfolios', 'forecasting market trends', and 'conducting sentiment analysis' had positive impact on the investment decision. The 'financial literacy' positively moderates the relationship between forecasting market trends and investment decisions. It negatively moderates the relationship between optimizing portfolios and investment decisions.

Nametala et al. (2023) differentiates from the other studies with proposing an investor bot to identify profitable opportunities by continuous re-optimization. The investor bot selected the easily traded assets to compose its investment portfolio in the first step. Then, the asset which had the biggest share in the portfolio was selected. The predictor module generated increase, decrease, and/or stable signals according to the predictions by SMA, ARMA, ARIMA, SARIMA, GARCH, and MLP. Then, the negotiator module evaluated the prediction module signals and calculated the performance metrics of the available investment strategy. The 'Brazilian stock market data' and 'taxes imposed by Brazilian government' were used to test investor bot performance. The investor bot constructed the portfolio with the assets from different market segments such as banks, metallurgical, mining, natural gas, and construction companies. The bot negotiated 70.5% of the opportunities and predicted 41.1% of the operations correctly. It performed better than the best individual setups of SMA, SARIMA (and simplifications), GARCH, MLP ANNs, Bovespa index, and Buy and Hold.

3.3.2 Forecasting Examples with Chatbots

From the points of Leigh (2024) and Hochmair et al. (2024), we proposed examples to see the macroeconomic indicator forecasting performance of the chatbots. In our study, we provided the annually time series of GDP current prices (from 1995 to 2010) and unemployment rate (from 2000 to 2010) for EU to the ChatGPT 5.2 and Gemini AI Pro. In our first prompt, we requested to use exponential smoothing (ES) and moving average (MA) methods in these two variables forecasting from 2011 to 2024. To evaluate the forecasting performances of the chatbots, the MAPE of the processes were also requested. The forecasted values for annual GDP current prices by both chatbots are given in Table 2. Gemini AI Pro applied 3-year moving average (MA(3) and simple exponential smoothing (SES) method (smoothing constant $=\alpha=0.9$). The MAPE of the predictions with MA(3) and SES($\alpha=0.9$) are 6.23% and 4.12%, respectively. For ChatGPT 5.2, the MAPE statistics of the MA(3) and SES are 6.23% and 3.84%. The MA(3) forecasting performances of the chatbots are the same. However, the performance of the ChatGPT 5.2 in the smoothing constant optimization is better than the Gemini AI Pro. For this reason, the MAPE of the ChatGPT 5.2 forecasting is less than Gemini AI Pro in SES method. It can be concluded that the ChatGPT 5.2 shows better performances than Gemini AI Pro.

The forecasted values for the unemployment rate by both chatbots are given in Table 3. The ChatGPT 5.2 and Gemini AI Pro applied the MA(3) and SES ($\alpha\approx1$)

methods. While the MAPE of the MA(3) is 12.15%, the MAPE of the SES($\alpha \approx 1$) method is 6.76% for both chatbots. Therefore, the ChatGPT 5.2 and Gemini AI Pro show the same performance in predicting unemployment rate.

In MA(3) and SES(α) methods, the one-step-ahead rolling forecasts were requested from chatbots. Rolling one-step-ahead evaluation is partially captured the structural changes in time series. However, it cannot capture the strong upward or downward trend. This situation produces relatively large errors compared with trend-adjusted methods. For this reason, we write a second prompt which is not indicated the forecasting methodology. In this prompt, The ChatGPT 5.2 and Gemini AI Pro are free to use any model they want to predict the annual GDP current prices/unemployment rate with the aim of minimizing MAPE.

For GDP current prices forecasting, the ChatGPT 5.2 tested Holt Linear Trend, Holt Damped Trend, and ARIMA(1,1,1) models. Since the GDP current price series behave almost like a smooth deterministic trend over 1995–2019, the Holt's linear trend model is found a well-suited forecasting model by the chatbot (Table 2). This model considers both systematic variation (level) and random variation (trend) in time series forecasting (Meade, 2000). Therefore, it can overcome data fluctuations by continuously updating the level and trend estimates (Ariyanto & Nugraha 2024). The model shows better performance with 3.85% MAPE statistic. Gemini AI Pro tested several time-series models such as linear regression and polynomial trend. Since the Holt's Linear Exponential Smoothing model produced the lowest MAPE (4.20%), the Gemini AI Pro also selected the same model like ChatGPT 5.2. However, the ChatGPT 5.2 shows slightly better performance than Gemini AI Pro according to MAPE. When the performance of the models compared with the forecasting models in our first prompt, the ChatGPT 5.2 shows the same performance in model free prompt. The Gemini AI Pro selected model performance is slightly worse than the forecasting models in the first prompt.

After testing Holt Linear Trend, Holt Damped Trend, ARIMA(1,1,1), Linear Trend Regression, and Quadratic Trend Regression models, the ChatGPT 5.2 selected Linear Trend Regression method for unemployment rate forecasting. The forecasted values for each year were given in Table 3. The MAPE was calculated 16.86%. The Gemini AI Pro selected Naive forecasting model. The forecasted value for the next year is simply the value of the current year in this model (Wu et al., 2021; Kirana & Bhawiyuga, 2021). Since the unemployment rate exhibits high volatility and random walk characteristics, Naive forecasting model produces the minimum MAPE compared with the complex smoothing models. The MAPE of the Naive forecasting model was calculated 6.76%. Therefore, the Gemini AI Pro shows better performance than ChatGPT 5.2 according to MAPE statistic. However, both chatbots did not show good performance in selecting forecasting model for unemployment rate. For Gemini AI Pro, the MAPE statistics in the first and second prompts are equal. For ChatGPT 5.2, the MAPE of model free prompt were worse than the MAPE statistics in the first prompt.

While advanced artificial intelligence (AI) models are increasingly applied in macroeconomic forecasting, the ChatGPT 5.2 and Gemini AI Pro did not select any advanced AI model in forecasting GDP current prices and unemployment rate. Furthermore, these models were not evaluated by chatbots during the selection of the optimal forecasting method. A third prompt was written to the chatbots concerning this

Table 2 The GDP current prices forecasted by Gemini AI Pro and ChatGPT 5.2

Years		The Results of the Gemini AI Pro			The Results of the ChatGPT5.2		
	Real Values	Predicted Values with MA(3)	Predicted Values with ES (α=0.9)	Predicted Values with Holt's Linear ES	Predicted Values with MA(3)	Predicted Values with SES	Predicted Values with Holt's Linear ES
2011	2,832,848	2,720,239.64	2,734,530.84	2,836,654.51	2,720,239.64	2,742,499.49	2,822,741.84
2012	2,850,179	2,741,015.50	2,823,016.62	2,918,879.31	2,741,015.50	2,832,396.63	2,902,582.98
2013	2,883,617	2,808,667.53	2,847,462.81	3,001,104.11	2,808,667.53	2,850,090.14	2,982,424.13
2014	2,951,282	2,855,547.98	2,880,001.15	3,083,328.91	2,855,547.98	2,883,448.89	3,062,265.28
2015	3,056,000	2,895,025.88	2,944,153.98	3,165,553.71	2,895,025.88	2,950,942.91	3,142,106.42
2016	3,137,766	2,963,632.85	3,044,815.35	3,247,778.51	2,963,632.85	3,055,474.66	3,221,947.57
2017	3,274,013	3,048,349.42	3,128,471.14	3,330,003.31	3,048,349.42	3,137,354.77	3,301,788.72
2018	3,386,792	3,155,926.43	3,259,458.90	3,412,228.11	3,155,926.42	3,273,329.81	3,381,629.87
2019	3,511,284	3,266,190.41	3,374,058.60	3,494,452.91	3,266,190.41	3,386,224.59	3,461,471.01
2020	3,371,066	3,390,696.17	3,497,561.01	3,576,677.71	3,390,696.17	3,510,658.21	3,541,312.16
2021	3,665,898	3,423,047.23	3,383,715.77	3,658,902.51	3,423,047.23	3,371,764.26	3,621,153.31
2022	3,992,185	3,516,082.76	3,637,680.20	3,741,127.31	3,516,082.76	3,664,427.80	3,700,994.45
2023	4,257,321	3,676,383.32	3,956,734.68	3,823,352.11	3,676,383.32	3,990,546.39	3,780,835.60
2024	4,376,645	3,971,801.62	4,227,262.55	3,905,576.91	3,971,801.62	4,255,987.33	3,860,676.75

Table 3 The unemployment rate forecasted by Gemini AI Pro and ChatGPT 5.2

Years		The Results of the Gemini AI Pro			The Results of the ChatGPT5.2		
	Real Values	Predicted Values with MA(3)	Predicted Values with ES ($\alpha \approx 1.0$)	Predicted Values with Naïve Model	Predicted Values with MA(3)	Predicted Values with SES	Predicted Values with Linear Trend Regression
2011	10.08	8.90	10.07	10.07	8.90	10.07	8.64
2012	11.06	9.82	10.08	10.08	9.82	10.08	8.55
2013	11.58	10.40	11.06	11.06	10.40	11.06	8.46
2014	11.00	10.91	11.58	11.58	10.91	11.58	8.37
2015	10.18	11.21	11.00	11.00	11.21	11.00	8.28
2016	9.25	10.92	10.18	10.18	10.92	10.18	8.19
2017	8.28	10.14	9.25	9.25	10.14	9.25	8.09
2018	7.39	9.24	8.28	8.28	9.24	8.28	8.00
2019	6.79	8.31	7.39	7.39	8.31	7.39	7.91
2020	7.23	7.49	6.79	6.79	7.49	6.79	7.82
2021	7.05	7.14	7.23	7.23	7.14	7.23	7.73
2022	6.15	7.02	7.05	7.05	7.02	7.05	7.64
2023	6.04	6.81	6.15	6.15	6.81	6.15	7.55
2024	6.03	6.41	6.04	6.04	6.41	6.04	7.46

Table 4 Statistical Significance Test for Best Performed Models

Chatbots	Variables and Models	Bias Tests	Autocorrelation Tests
Gemini AI Pro	GDP Current Prices SES Model	p-value = 0.0013	Durbin Watson Statistics = 0.652
		significant positive bias	strong positive autocorrelation
	Unemployment Rate SES Model	p-value = 0.0963	Durbin Watson Statistics = 0.853
		no bias	strong positive autocorrelation
ChatGPT 5.2	GDP Current Prices SES Model	p-value = 0.0028	Ljung–Box test (lag 5): p-value = 0.8805
		systemic bias	No autocorrelation
	Unemployment Rate SES Model	p-value = 0.108	Ljung–Box test (lag 5): p-value = 0.127
		no bias	No autocorrelation

issue. Since the advanced AI models prediction performances are critically dependent on data volume and structural complexity, they were not favored in our forecasting scenarios by the chatbots. Another important point in the responses of chatbots was that advanced AI models are prone to overfitting for small-sample and low-frequency datasets. Consequently, these models were not tested or selected by chatbots as they would not perform better than simple statistical models for the datasets in our example.

Gemini AI Pro shows best forecasting performance in GDP current prices with SES ($\alpha = 0.9$) model and unemployment rate with SES ($\alpha \approx 1$) model. ChatGPT 5.2 shows best forecasting performance in both GDP current prices and unemployment rate with SES model. To evaluate the forecasting performance of the chatbots, the statistical significance test and the model stability tests were also requested from the chatbots for these best performed models. The results were given on the Table 4.

For bias assessments, both chatbots selected t-test. The very low p-values which are equal to 0.0012 in Gemini AI Pro calculation, 0.0028 in ChatGPT 5.2 calculation indicate a statistically significant bias in GDP current prices prediction. It can be originated from the post-2020 growth acceleration. This acceleration was not captured by the historical data forecasting. The ChatGPT 5.2 and Gemini AI Pro failed to reject the null hypothesis of zero bias at the standard 5% level for unemployment rate forecasting. This unbiased situation indicated that the SES model is statistically more robust.

While Gemini AI Pro selected Durbin Watson (DW) test to consider autocorrelation in the model, the ChatGPT 5.2 selected Ljung–Box tests. The Ljung–Box tests indicated that forecast errors are not serially correlated for either GDP current values or unemployment rate models. However, the DW statistics are below 2.0 for forecasting models of GDP current prices and unemployment rate. This indicates strong positive autocorrelation in the residuals. Therefore, the forecasting models are considered as trend-followers rather than trend-predictors for both variables.

4 Conclusions

This study provides the literature review on AI tools and chatbots usage in forecasting and/or nowcasting of economic and financial indicators. The WoS database is searched with the combination of “forecasting”, “nowcasting”, “chatbot”, “artificial intelligence”, “bot,” and “chat” keywords. The initial research is concluded 399 papers.

After the screening and abstract evaluation, the 111 papers are selected for reviewing. The full papers of all studies are reviewed and classified according to research topic. The top seven research area are 'stock market index (23 studies)', 'oil and electricity prices (11 studies)', 'cryptocurrency (6 studies)', 'exchange rates (6 studies)', 'bankruptcy (6 studies)', 'unemployment rate (4 studies)', and 'GDP (5 studies)'.

For the stock market research area, the most of the studies were investigated the Down Jones, S&P500 and Borsa Istanbul stock markets. The deep learning methods such as LSTM, CNN, RNN, XGBoost enable better modeling of complex and non-linear relationships in financial markets than traditional time series models such as ARIMA and GARCH. The use of sentiment data (Ballesteros & Miranda, 2024) and visual data of stock chart images (Park et al., 2024) have stood out as new and effective approaches.

The high accuracy metrics of the hybrid models draw attentions in the electricity and oil prices area. The studies considered the energy prices for different countries such as Austria, Ukraine, UK, French, and China. In addition to weather conditions, socio-economic variables, energy demand, and Google Trend data also used as input variable to track the energy consumption. For this research area, the consideration of the greenhouse gas emissions and the alternative energy sources can be proposed as new research topics.

For the cryptocurrency field, the deep learning models, such as LSTM and GRU, demonstrated high accuracy in predicting Ethereum, Bitcoin and Ripple prices (Demirci & Karaatli, 2023; Ali et al., 2024). Hybrid models which combine ANN with metaheuristic algorithms outperformed traditional optimization methods with achieving lower error rates (Behera et al., 2023).

The exchange rate field included the variety of currencies such as Euro, US dollar, Canadian dollar, British pound, Swiss franc, and Mauritian currencies. The most of the studies concluded that the performances of the NNs outperformed the linear time series and autoregressive models. However, selecting appropriate architecture and algorithms for NNs are still important concerns in this research area. Therefore, hybrid methodologies which contributed to optimization of model parameters such as GA and PSO can be used for future studies.

The studies on bankruptcy field used the classification techniques for defining both personal and market bankruptcies in several nations, including the US, Spain, Slovakia, and Taiwan. The decision tree models, SVM, LightGBM, and fuzzy set approach with ANN models consistently showed high accuracies in classification. There were greater variations in the number of input variables in these studies. Therefore, the PCA or lasso models can be used for dimensionality reduction. The key independent variables can also be identified using the multi-criteria decision making models.

The nowcasting applications based on the Twitter data and Google Trend data draw attention in the unemployment rate field. While nowcasting errors ranges from 27.7 to 0.018 (RMSE), the forecasting errors ranges from 0.20 to 0.31 (RMSE). These prediction performances mean that the input data in qualitative nature showed better performance in predicting unemployment rates.

The economic effects of Covid19 have been extensively included by the studies in GDP field. Since the quarterly publication frame of the GDP values, the forecasting or nowcasting efforts are expected to high. However, the number of studies in this area is not line with this expectation.

The most of the AI applications in our review pool belong to the field of ML and deep learning. The widely used methods are ANN (Mutascu & Hegerty, 2023; Kreiner & Duca, 2020; Letkovsky et al., 2023), LSTM (Alfarra & Hagag, 2024; Ali et al., 2024; Demirci & Karaatli, 2023; Guo et al., 2023; Grybauskas et al., 2023; Metlek, 2022; Al-Khasawneh et al., 2024), GRU (Demirci & Karaatli, 2023; Guo et al., 2023) and RNN (Guo et al., 2023; Metlek, 2022) in forecasting applications. The performance of these AI methods compared with the traditional ARIMA (Alfarra & Hagag, 2024; Armagan, 2023a) and GARCH (Nag & Mitra 2002; Amelot et al., 2021; Al-Khasawneh et al., 2024).

While NN models shows promising results in forecasting and nowcasting application, the major problem is about to select configuration parameters such as 'the number of hidden layers', 'nodes in each hidden layer', and 'input nodes' (Nag & Mitra 2002). To optimize the configuration parameters in NNs and provide clean data to model, the hybrid models are constructed with PSO (Cohen, 2023; Jursa & Rohrig, 2008), bayesian algorithm (Diallo et al., 2023; Moreira et al., 2024), and RF algorithm (Ali et al., 2024; Ekinci & Sen, 2024; Antoniadis et al., 2024). Since the learning and training phases of the hybrid models can be time consuming (Li et al., 2022; Armagan, 2023a), the lasso regression models (Li et al., 2022; Bredin et al., 2021), granger causality (Yu et al., 2019; Kraaijeveld & De Smedth, 2020) and PCA (Hu & Ansell, 2009; Kreiner & Duca, 2020) are also used for dimensionality reduction.

The forecasting and nowcasting models are constructed by using Matlab package (Zhang & Zhang, 2023; Moreno, 2009; Korol & Fodadis, 2022; Sencan & Sencan Sahin, 2022), Python package (Antony & Natarajan, 2024; Grybauskas et al., 2023, Armagan, 2023a; Agakishiev et al., 2025; Brygala & Korol, 2024), and R programme language (Navarro- Galera et al., 2024; Aznarte et al., 2011), generally. In addition, the Azure Machine Learning Studio web solution is proposed by Ballesteros & Miranda (2024). This package enables the acceleration and management of ML project using open source platforms such as PyTorch, TensorFlow, or Scikit-learn.

The most used performance indicators are MSE, RMSE, MAE and forecasting accuracy rate R^2 (Jiang et al., 2023; Nag & Mitra, 2002; Yilmaz & Yildiztepe, 2024; Behera et al., 2023; Ding & Zhang, 2023; Moreno, 2009; Antoniadis et al., 2024; Guo et al., 2023; Egrioglu, 2014; Li & Tang, 2020; Al-Khasawneh et al., 2024; Ballesteros & Miranda, 2024, Sencan & Sencan Sahin, 2022; Park et al., 2024). In classification problems, the type I error and type II error (Korol, 2013; Ekinci & Sen, 2024; Brygala & Korol, 2024) are generally used as performance indicators.

The number of the nowcasting applications is less than the forecasting applications because of the limitation related to real time data availability. The tweets gathered from Twitter (Lehrer et al., 2021; Kraaijeveld & De Smedt, 2020; Nia et al., 2022) or search queries on Google Trend (Yu et al., 2019; Grybauskas et al., 2023) are generally used for nowcasting applications. The users of these platforms are belonging to certain subset of the population which are mobile users, urban residents and have internet connectivity (Nia et al., 2022). Therefore, the representation of this subset to whole population is doubtable. Other major limitation about the Twitter data is the hashtags. Twitter users added many hashtags, which do not relate to tweet content, to attract attention (Kraaijeveld & De Smedth, 2020). This situation reduces the quality of the dataset. For

Google Trend data, the limitations are difficulty in topic-related keywords selection (Yu et al., 2019) and timing sensitivity of the keywords (Yu et al., 2019; Nia et al., 2022).

The literature about the chatbot usage in economy is limited by conceptual studies. This finding indicated that, the forecasting and nowcasting applications with chatbots are still in progressing. The focus on the conceptual analysis does not reach the maturity level. The number of studies indicated that, the forecasting and nowcasting application with chatbot technologies have not been adequately studied. To show the forecasting performance of the Gemini AI Pro and ChatGPT 5.2, we employ forecasting examples with the GDP current prices and unemployment rate in EU. The MAPE statistics calculated by Gemini AI Pro for GDP current price forecasts are 6.23% (MA(3)), 4.12% (SES), and 4.20% (Holt's linear trend), respectively. The MAPE values calculated by ChatGPT 5.2 are 6.23% (MA(3)), 3.84% (SES), and 3.85% (Holt's linear trend). The best prediction performance was shown by the ChatGPT 5.2. For unemployment rate forecasts, the MAPE values calculated by Gemini AI Pro are 12.15% (MA(3)), 6.76% (SES), and 6.76% (Naive Forecasting Model), respectively. The MAPE values calculated by ChatGPT 5.2 are 12.15% (MA(3)), 6.76% (SES), and 16.86% (Linear Trend Regression). Both ChatGPT 5.2 and Gemini AI Pro showed good performances in predicting unemployment rate forecasting.

The advantages of AI and chatbot technologies in forecasting and nowcasting applications can be listed as 'analyzing higher volume of both structured and unstructured data', 'providing high accuracy rates', 'addressing the potential shortcomings of traditional forecasting and nowcasting methods', 'identifying and responding to risks such as political instability, natural disasters, and economic fluctuations' (Zhang, 2024). Despite the advantages, these technologies also come with limitations (Ozturk, 2024; Fernandez, 2019). The main risks stem from the potential bias in the results obtained using these tools. According to findings of Nguyen (2024), nearly, all chatbots forecasts showed statistically significant anchoring in forecasting exchange rates. The 'chain of though' and 'ignore previous' approaches failed in minimizing the anchoring effect especially in GPT-3.5 forecasts in this study. Therefore, the users of AI assistant in financial decision making should consider the potential risk associated with the bias.

In addition to data bias and anchoring effects, there are several additional limitations inherent to the fundamental nature of large language models (LLMs). One of these limitations is that LLMs lack structural and causal understanding of economic systems. They understand contextual relationships (Wang et al., 2024; Stevens, 2024) rather than causal inferences. While the outputs of the prediction model may be statistically consistent, they may be economically inconsistent. Another limitation is sensitivity to structural breaks. Chatbots operate under the assumption of the continuity of historical patterns (Eyinade et al., 2025). In other words, they assume that the future will resemble the past. Therefore, they may exhibit poor prediction performance under structural breaks or unprecedented shocks (Eyinade et al., 2025).

Another point to consider when making predictions with chatbots is their sensitivity to prompt contents (Stevens, 2024). Small changes in the prompt can alter model selection or parameterization of the chatbot. Therefore, the outputs of the prediction processes are affected by this situation. Additionally, limited methodological transparency (Eyinade et al., 2025) and the potential hallucination of technical details

(Stevens, 2024; Hassani & Silva, 2024) are other constraints which reduce the reliability of chatbot predictions.

Despite all these limitations, chatbots are good decision support tools. Since they are technologies open to development (Hassani & Silva, 2024), these limitations frequently encountered in the free versions are not often seen in the versions offered with payment plans. Therefore, these limitations do not indicate that LLMs have no place in time series analysis. AI and chatbot technologies can be applied to forecasting studies for any sector. For the field of economy, NN models with optimization algorithms provide better prediction performances. In addition to that, using input data in qualitative nature also contribute to the performance of the models. The forecasting and nowcasting studies with chatbot technologies have not been adequately addressed in literature studies. The efforts on this direction can be stood out as new and effective approaches for the forecasting and nowcasting applications.

Our research enriches the existing literature in proposing practical guide on the AI and chatbot technologies used for forecasting and nowcasting applications in economy. However, these applications should be supported with the AI strategies of governments. The most of the institutions which implement conversational chatbots were found in the US, followed by UK, China, and Germany (Lin et al., 2023). This can be originated from 'the technological infrastructure', 'the number of skilled professional in related field', and 'the availability of data'. Authorities have begun to incorporate AI technologies into some of their activities for operating efficiency and cost savings. For instance, the central bank of Italy is using AI techniques to predict price moves on the real estate market. The central bank of the Netherlands has also used AI techniques to anticipate potential liquidity problems at financial institutions (Fernandez, 2019). However, the divergence between countries came into the question. According to European Parliament (2024), while almost €32.5 billion was invested in EU AI companies, more than €120 billion was invested in US AI companies between 2018 and the third quarter of 2023. The problems originated from this diversity can be handled by the collaboration especially in 'harmonizing data protection laws' and 'establishment of moral AI standards'.

Despite the authoritative nature of the WoS database, our study does not cover all the literature on AI and chatbot technologies in forecasting and nowcasting applications in economy. It is limited with the WoS database. Since the AI and chatbot technologies are hot topic in recent years, the effect of time on the amount of literature is also crucial. With the use of alternative databases and keywords, the content of this review can be expanded. The scope of this study limited with the technical applications of AI and chatbot technologies in macroeconomic indicators forecasting and nowcasting. Therefore, the theoretical contribution of this review is limited with the AI and chatbot technologies background. The forecasting performances of Gemini AI Pro and ChatGPT 5.2, which are offered with a payment plan, are presented in our study. The forecasting applications can be repeated with using other chatbots such as Copilot and Bing AI. When we request forecasting applications with nonlinear methods from chatbots, they propose the Python codes of the methods. These codes can be used in future studies to investigate the performance of the chatbots in writing code.

Appendix

Table 5 The Summary of the Selected Studies in Review Pool

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Antony and Natarajan (2024)	Bond Yield Spread	Forecasting Related Study: Auto-Regressive Integrated Moving Average (ARIMA) Linear Regression (LR) Deep Recurrent Neural Network-Long Short-Term Memory (DLSTM)	The DLSTM models most fairly predict the price of government bonds over both the short and longer horizons when compared to ARIMA and LR. The deep LSTM neural network outperformed LR by 65.74% (MAPE) and ARIMA model by 99.44% (MAPE) in 10-year bond case forecasting. For the 3-year bond case forecasting, the deep LSTM neural network outperformed LR by 89.66% (MAPE) and ARIMA by 99.86% (MAPE).
Armagan (2023b)	Stock Index	Forecasting Related Study: Facebook Prophet Model (FPM) Convolutional Neural Network (CNN)	The deep learning models showed better performance than the traditional model. The performance indicators for ARIMA model were 25% for MAPE and 0.30 for R^2. The performance indicator of FPM method were 34.44% for MAPE and 0.62 for R^2 in one-year forecast obtained by training. The CNN model performed better than ARIMA and FPM with 7.60% for MAPE and 0.95 for R^2.
Beltrametti et al. (1997)	Foreign Exchange Market	Forecasting Related Study: Artificial Agent Vector Auto Regression (VAR)	The artificial agent bets on the exchange rate going up or down according to its observation on foreign exchange markets and exchange rate determinants. The agent captured the fundamental trends. It took the right actions with about 70% frequencies. While the performance of the agent was slightly better than VAR model for dollar-mark exchanges, it outperformed the VAR model for dollar-yen market.
Bredin et al. (2021)	Crude Oil Prices	Forecasting Related Study: LASSO Regression Model	The NS factor formulation provided the best prediction for medium length holding periods. The significances of the coefficients of the NS factors were robust to the inclusion of macroeconomic and oil market related variables. The lasso model with NS factors decreased forecast errors in out of sample prediction for all horizons.
Cortazar et al. (2021)	Crude Oil Prices	Proposed New Model: Multifactor Stochastic Pricing Model	The results of the multifactor stochastic pricing model indicated that the analyst forecasts were valuable for predicting future energy prices. The relative MSPE was used as the success ratio metric. For all forecasting horizons, the proposed model was better than both the No-Change forecast and the Bloomberg's Consensus Model.

Table 5 (continued)

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Cotter et al. (2023)	Asset Price	Forecasting Related Study: Simple averaging schemes: mean, trimmed mean, median, and weighted-mean forecasts. Performance-based combination forecast- discounted mean squared forecast error (DMSFE)	The combination forecasts of commodity returns outperform the benchmark historical average return forecasting. They generate positive and statistically significant out-of-sample in the range 0.81%–3.77%. The results also establish an important implication of the positive sign of the risk premia. It associates with the state variables that predict an increase in future stock market returns and economic activity and a decline in stock market volatility.
Diallo et al. (2023)	Industries' Portfolio Returns	Proposed New Model: Fama–French Three and Five Factor Models (with Bayesian optimization-support vector regression (BSVR) approach)	The Fama–French BSVR three-factor model outperformed the Fama–French BSVR five-factor model. The Fama–French BSVR three-factor estimations attained correlation coefficients of 94% for portfolio returns for the consumption and manufacturing industries, 92% for the high-tech industry, 91% for the mining, construction, transportation, hotels, entertainment, and finance industries. However, for the Fama- French BSVR five-factor model, the correlation coefficients lie between 48% (health industry) and 89% (high-tech industry).
Ding and Zhang (2023)	Carbon Emissions	Proposed New Model: New-Information-based Grey Model (integrated inventive damping accumulative generating operator, data smoothing index, and particle swarm optimization)	The average MAPE values of the proposed model for 30 provinces are less than 5% for in-sample and 10% for out-sample periods. This model can strikingly improve the forecasting ability. It also significantly presents the universal excellent prediction performance to provincial CO2 emission time series.
Egrioglu (2014)	Stock Exchange Composite Index	Proposed New Model: Fuzzy Time Series Method based on Particle Swarm Optimization	The method was applied to Index 100 for stocks and bonds exchange market of Istanbul (IMKB), Taiwan future exchange (TAFEIX), and Taiwan stock exchange capitalization weighted stock index (TAEIX). According to RMSE indicator, the proposed model outperformed the other models in TAIEX data set. For IMKB and TAIFEX data set, the proposed model also showed better performance especially for five fuzzy sets and second order model situations.

Table 5 (continued)

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Ekinci and Erdal (2017)	Bank Failure	Forecasting Related Study: Base learners- Logistic, J48 and Voted Perceptron Ensemble- Multiboost-j48 (M-J48), multiboost-voted perceptron (M-VP), multiboost- logistic (M-L), bagging-j48 (B-J48), bagging-logistic (B-L), bagging-voted perceptron (B-VP)) Hybrid Ensemble- Random sub-space-bagging-j48 (RS-B-J48), random subspace-bagging-logistic (RSB-L), random subspace-bag-ging-voted perceptron (RS-B-VP), random subspacemultiboosting-j48 (RS-M-J48), random subspace-multiboosting-logistic (RS-M-L)	Experimental results indicate a clear outperformance of hybrid ensemble machine learning models over conventional base and ensemble models. There is usually a trade-off between the sensitivity and specificity. Thus, best models reach the low specificity rates. While B-L and RS-B-J48 models have 0.196, the M-J48 and RS-M-J48 models have 0.219 specificity rates. As expected, the lowest average specificity rate is belongs to hybrid ensembles with 0.220. B-L and RS-B-J48 models generate the best classification rates with 83.783. Hybrid ensembles have the best average classification rate with 78.378.
Ekinci and Sen (2024)	Bank Failure	Forecasting Related Study: J48, Logistic Regression, Multi-layer Perceptron, Random Forest, Extreme Gradient Boosting (XG-Boost), and Cost-Sensitive Forest (CSForest)	Logistic Regression has the smallest false-negative rate with 18.64%. The second-best model is the CSForest model with a 20.34% false-negative rate. The XGBoost as the worst model has turned 20 "false negatives". The CSForest model would be the best model with 99.72% true positive rate.
Garcia et al. (2023)	Stock Market Index	Proposed New Model: Supervised Machine Learning Early Classification Technique	A novel supervised ML early classification technique provided both binary classification (rise/fall situations) and triple ranking (go up more than 0.5%/go down more than 0.5%/remain between up and down range). For all indicators and prices, the accurate rate of binary classification was above 50% while the triple classification was above 33.33%.

Table 5 (continued)

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Gazely et al. (2000)	Inflation	Forecasting Related Study: Artificial Neural Network	The exogenous input variables were the monetary indices (simple sum or Divisia), the inflation model used `Model 1' or `Model 2' (where the latter includes an autoregressive term) and the absence or presence of a time trend (`No' or `Yes'). Divisia monetary aggregates are found to outperform their simple sum counterparts in forecasting potential. RMSEs for inflation in the USA are found to be less than half of those the UK across the board. In the case of Italy, the results are in line with the USA and UK.
Ghoujdam et al. (2024)	Credit Risk	Forecasting Related Study: Logistic Regression Artificial Neural Network Decision Tree	The decision tree shows best performances with 89.4% accuracy and recall at 89.96%. According to F1- score and the AUC, the neural networks model outperforms the decision tree and logistic regression. For all metrics, the logistic regression model shows the worst performance (64.23% accuracy, recall at 61.62%, 64.5 F-1 score, and 72.9% in AUC) in predicting credit risk.
Guo et al. (2023)	Crude Oil Spot Prices	Forecasting Related Study: Recurrent Neural Networks (RNN), Long-Short Term Memory (LSTM), Gated Recurrent Unit (GRU), Support Vector Regression (SVR), Multilayer Perceptron (MLP), Convolutional Neural Network (CNN) and Back Propagation (BP).	When all the influencing factors were considered, the GRU model for 5-days-step had the smallest evaluation indicators. The MAE, MSE, RMSE, and MAPE statistics were 1.8341, 0.0594, 2.4367, and 2.4757, respectively. Considering all influencing factors at the same time for 10-day steps, the GRU model maintained the lowest evaluation indicators. The MAE, MSE, RMSE, and MAPE were 1.8634, 0.0614, 2.4781, and 2.5019, respectively.
Jankova and Dostal (2022)	Stock Index	Proposed New Model: Interval Type-2 Fuzzy Logic Model	The six experimental models were constructed with different degrees of uncertainty which are no uncertainty, 10% uncertainty, 20% uncertainty, 30% uncertainty, 40% uncertainty and 50% uncertainty. The 10% uncertainty model showed better performance according to 0.067 MSE and 0.075 MAE. The models which are included the most uncertainties with 40% and 60% showed the worst prediction errors. The best fuzzy model showed the similar performance with NNs in respect to 0.0075 MSE. It was significantly better than NNs according to MAE. Therefore, the proposed model competed with the standard models in the appropriate integration of uncertainty situations.
Jursa and Rohrig (2008)	Power Output of Wind Farm	Proposed New Model: Particle Swarm Optimization and Differential Evolution	With using the optimization algorithms, the prediction errors were reduced. The mean improvement in the prediction error of ANN specified by particle swarm optimization is 9.6%. The improvement of the manually specified neural network is 6.8%.

Table 5 (continued)

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Kaboudan (1999)	Stock Returns	Forecasting Related Study: Genetic Programming	The new measurement of corresponded to the ratio of the fitness value of subject series to the randomly shuffled of the same series. While "zero value" of the index indicated the unpredictable process, the "100" indicated that the process was totally predictable. The new measurement values for 30-minute data was close to zero. Therefore, the processes with 30-minutes data frequency were unpredictable. For one-minute data, only AT&T was found as predictable.
Kim and Chun (1998)	Stock Price Index	Proposed New Model: Probabilistic Neural Network (PNN)	They fractional changes of the Singapore Stock Price Index from Level 0 to Level 6. The proposed model showed best performance, especially for Level 3. It also outperformed the recurrent neural networks (RNN) and back propagation network (BPN) models for Level 4 according to type I (0.616) and type II (0.204) errors.
Lehrer et al. (2021)	Social Media Sentiment Index	Proposed New Model: Heterogeneous Autoregressive Realized Volatility Model (HAR) Model Averaging with Heterogeneous Autoregressive Realized Volatility Model (MAHAR)	Including Twitter sentiment in model can significantly improve the forecasting accuracy of The Chicago Board Options Exchange Volatility Index (VIX), particularly in short time horizons. Using a HAR estimator provides large gains in predictive accuracy. The MAHAR estimator has greater forecast accuracy than other HAR-type estimators.
Li and Paraschiv (2022)	Wind and Solar Power Infeed	Forecasting Related Study: Stochastic models- Ornstein Uhlenbeck (OU), Chan, Karolyi, Longstaff and Sanders (CKLS), and Cox-Ingersoll-Ross (CIR) Probabilistic model- Gaussian Mixture Models (GMM)	The GMM is a reliable tool to simulate the updates in the weather data. It shows superior simulation performances versus the classical stochastic models OU, CKLS and CIR. For the out-of-sample analysis, the proposed models have different prediction performances according to the weather forecast updates frequency. The GMM performs better than the stochastic models during the low-frequency-update period. In the high-frequency-update period, the stochastic models show superior performance.
Li and Tang (2020)	Stock Exchange Composite Index	Proposed New Model: Wavelet transform + Filter Cycle Decomposition + Multi Lag Neural Networks (WT-FCD-MLGRU)	The performance of proposed method was compared to ARIMA and SVR according to MAE, RMSE, MAPE, and SMAPE indicators. The proposed model performed better than both ARIMA and SVR methods. Especially, the proposed model showed highest accuracy in predicting 30 days' daily prices of S&P500, IXIC, DJI and SSE.
Li et al. (2022)	Financial Risk Detection	Forecasting Related Study: Case-Based Reasoning (CBR)	The proposed CBR method has a relatively superior prediction performance compared to the widely used classification ML methods. When compared to other black-box machine learning methods, the CBR system is capable of interpreting the result of the financial risk prediction. Further, it detects the schemes to decrease or avoid financial risk.

Table 5 (continued)

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Lo and Singh (2023)	Risk Premia	Forecasting Related Study: Local Interpretable Model-agnostic Explanation (LIME)	The proposed approach captures the interaction between factors and risk premia. In other words, the proposed deep neural networks outperform linear ML models. It can be extended to predict returns at multiple horizons. It also offers a robust framework that can be applied in various applications for portfolio construction, performance attribution, and risk management.
Lu et al. (2022)	Stock Closing Price	Proposed New Model: Convolutional Neural Networks (CNN)+GRU +LSTM	The LSTM and GRU models showed the similar performances in predicting stock prices according to MAE, RMSE, and R^2 indicators. If the CNN part added to GRU algorithm, all errors decreased. With the attention mechanism, the MAE reduced by 2.5%, RMSE reduced by 1%, MAPE reduced by 2.3% while the R^2 increased by 0.0007.
Luo et al. (2019)	Business Operations: Stock Index Future (IF)	Proposed New Model: Deep Deterministic Policy Gradient based Convolutional Neutral Networks (CNNs)	The proposed AI-trader outperforms other methods with the use of real stock-index future data. The proposed AI-trader shows good performance in extracting features of financial markets and learning trading actions. In the test period, It obtains a promising annual yield. It also performs more stably and profitably compared with random trading strategy and the basic RRL system.
Maknickiene (2018)	Investor Sentiment Index	Forecasting Related Study: Evolino Recurrent Neural Networks	This paper hypothesis that dynamics in the market are unequivocally related to individual investors' sentiments. This relationship occurs when the sentiments are expressed strongly and exceed boundaries. The results of this research confirm the hypothesis. Since the sentiments data is less volatile than the financial indicators, the forecasting accuracy of the model increased to 87%.
Maknickiene (2020)	Forex Market: Exchange Rates Changes	Forecasting Related Study: Evolino Ensemble Prediction	The high and low distributions received by an Evolino ensemble allow the investor to increase the probability of success. This model can be successfully used to robotize trading in the currency market or to develop new Fintech services for investors.
Maknickiene and Maknickas (2013)	Portfolio Decision Making Process	Proposed New Model: Combination of Evolino Recurrent Neural Networks and Delphi Expert Evaluation Method	The proposed model increases the profitability of the building investment portfolios from 12 to 15% to 20–25% and 27–35% in 40 trading days with the different degrees of risk. Investments based on the knowledge of the prediction of the RNNs become more reliable and more profitable.
Moreira et al. (2024)	Interest rate (Selic)	Forecasting Related Study: Deep Neural Networks	The DNNs outperform statistical methods in the prediction process of the time series. The fine-tuning DNN hyper parameters lead to more accurate forecasting models without changing model size. The error rate above 0.00219 and training time above 84.28 s for DNN model.
Moreno (2009)	The monthly generation in hydraulic plant	Forecasting Related Study: Adaptive-Network-based Fuzzy Inference System (ANFIS)	In the MLR the value of MAPE statistic was 27.81%. With the ANFIS model, the training MAPE was 18.44% and validation MAPE was 12.87%. The most errors in ANFIS model are situated below 20% meanwhile most errors in MLR model are above that level.

Table 5 (continued)

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Onozó et al. (2024)	Sentiment Index	Proposed New Model: A Dictionary and Transformer-Based Sentiment Model	Transformer-based sentiment classifiers approach the accuracy of human annotators in the labeling process of the news as positive, negative and neutral. With active learning, the accuracy and robustness of these classifiers can be enhanced. The proposed sentiment indices show promising performances in nowcasting macroeconomic indicators according to Granger causality test and Dynamic Time Warping analysis.
Pinto and Marcal (2023)	Consumer Price Index	Proposed New Model: Reinforcement Learning Method	The proposed algorithm can improve forecast performance in a scenario which is structural change without losing significant performance under normal times. According to the consumer price index predictions for 38 countries, the RL mechanism was the best model in, approximately, 80% of the cases.
Qiu et al. (2021)	Fraud Risk Assessment	Proposed New Model: The Fraud Risk Assessment Model based on Multisource Evidence Theory	The proposed model is effective at evaluating the fraud risk of listed companies in China. The F1-measure value of the proposed model first increases and then decreases for each 5% increase in the sample size. However, the Dechow model increases always. The F1- measure value of the proposed model is always greater than the Dechow model. The proposed model has a higher precision, recall, and F-1 measure than the traditional probability model.
Qu and Yuan (2024)	Equity Premium	Proposed New Model: Momentum-Determined Indicator-Switching (N-MDIS) Strategy	The proposed strategy increase the equity premium out of sample predictions performance of MDIS from − 1.099% to 19.311%. The economic value of the N-MDIS strategy is also measured with CER gains. It increases gains from 0.397% to 3.127%. The robustness of the proposed new strategy is evaluated for different asset weight limitations and two different out of sample periods. These analyses also showed the superiority of the proposed strategy in predicting equity premium.
Ram et al. (2024)	Stock Market Index	Forecasting Related Study: Deep Deterministic Policy Gradient (DDPG) Proximal Policy Optimization (PPO) Recurrent Deterministic Policy Gradient (RDPG)	The static context and streaming context of the closing prices of Alphabet, Mackey-Glass, gold prices, and crude oil were evaluated. The performances of the algorithms were compared with Multi-Layer Perceptron (MLP), Support Vector Regression (SVR) and General Regression Neural Network (GRNN) methods. The DDPG algorithm showed better performance for static context. The GRNN showed the better performance for streaming context.
Ruan et al. (2020)	Investor Sentiment Index	Forecasting Related Study: Deep Learning Approach	The ISI is positively correlated with future stock market returns at a monthly frequency. However, It is negatively associated with subsequent returns over a longer horizon. Moreover, ISI outperforms other well-recognized predictors both in and out of sample. It can predict cross-sectional stock returns sorted by industry. While, the ISI is significantly positively associated with future aggregate dividend growth, it is not related to future dividend-price ratio.

Table 5 (continued)

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Salvagnin et al. (2024)	Dynamics of Carbon Emission Pricing	Proposed New Model: Gaussian Processes combined with Information Imbalance	The commodity-related variables such as the ERIX index are the most informative in Phase 3. The financial fluctuations related variables such as the uncertainty in the EUR/CHF exchange rate is the crucial determinant of the EU emission trading system in Phase 4. Using the top 3 most informative variables improves prediction performance, both for nowcasting and forecasting.
Schittenkopf et al. (2000)	Daily Closing Value of FTSE 099 index	Proposed New Model: Recurrent Mixture Density Network	Three different density specifications a standard GARCH model with a "normal distribution, heteroscedastic but neither skewed nor leptokurtic", a GARCH model with a "Student t-distribution, heteroscedastic, not skewed but leptokurtic", and a "recurrent mixture density network, heteroscedastic, skewed and leptokurtic" in a time dependent manner were compared. The empirical analysis demonstrates the importance of distributional assumptions in volatility prediction. The out of sample forecasting performance of NNs slightly dominates the GARCH models.
Sencan and Sencan Sahin (2022)	Gross Domestic Product	Forecasting Related Study: Adaptive Network-based Fuzzy Logic Inference System (ANFIS)	The exports, imports, government expenditures, consumer price index (CPI) and inflation rate were the inputs of the model. The model provided acceptable levels both for the MAPE (1.5137) and R^2 (0.9949) for Türkiye.
Sermpinis et al. (2016)	Stock Market Index	Proposed New Model: Adaptive Evolutionary Multilayer Perceptron (aDEMLP) Adaptive Evolutionary Wavelet Neural Network (aDEWNN)	The SPDR Dow Jones Industrial Average (DIA), iShares NYSE Composite Index Fund (NYC) and SPDR S&P 500 (SPY) exchange-traded funds (ETFs) were dependent variables in the analysis. The first fifteen autoregressive lag of the exchange rates (Euro/USD, USD/GBP, and USD/JPY), commodities such as gold bullion, silver, oil and copper, Nikkei 225 and FTSE 100 indexes are used independent variables. The proposed hybrid models performances are compared with traditional STAR, a moving average convergence/divergence model (MACD) and random walk model. The proposed hybrid models showed superior performance in predicting according to RMSE, MAE, Theil's U statistic, Pesaran–Timmermann (PT) test, and HLN statistics. These hybrid models also handled with data snooping bias and the ineffectiveness in parameter optimization process.
Sing et al. (2022)	Housing Prices	Proposed New Model: Artificial Intelligence Based Automated Valuation Models (AI-AVM)	The boosting AI-AVM models explain the 91.33% variation of public housing market price with 8.55% MAE. The 94.28% variation of the private housing market price is also explained by this model with 5.34% MAE.

Table 5 (continued)

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Stankev-iciene et al. (2017)	Portfolio Composition	Forecasting Related Study: Evolino Recurrent Neural Networks	Two methods of decision-making which are "close-to-close, based on one distribution of expected values" and "high–low based on the composition of two distributions" are proposed in the paper. Two strategies of portfolio selection were also compared: 1/N and optimized. Portfolios constructed using probabilities of high–low distributions are more profitable, effective and accurate than portfolios constructed using probabilities of close-to-close data.
Sun et al. (2023)	Inflation	Proposed New Model: Time-Varying Forward-Validation Model Averaging (TVFVMA)	TVFVMA consistently outperforms recursive estimation-based model averaging methods, including SAIC, SBIC, MMA and JMA. Thanks to the adaptive LASSO penalty used in TVFVMA, It has robust accurate forecasting ability. It may decrease forecast bias substantially more than increase estimation uncertainty.
Sun et al. (2023)	Financial Distress	Proposed New Model: Improved Decision-Directed Acyclic Graph (IDDAG) One-Versus-One (OVO) Decomposition Method (OVO-IDDAG)	The performance comparison is conducted for a variety of base classifiers and three imbalance processing mechanisms. The base classifier and imbalance processing mechanism are selected for the OVO-IDDAG model. It provides support for the practical application of OVO-IDDAG to MFDP. The OVO-IDDAG method overall outperformed the benchmark methods.
Wu et al. (2015)	Stock Exchange Composite Index	Proposed New Model: Autoregressive Moving Average (ARMA)+Support Vector Machine (SVM) Autoregressive Moving Average (ARMA)+ Probabilistic Neural Network (PNN) Back-propagation PNN model	Four types of agents were constructed to operate the average result of two quantitative models (ARMA+SVM, ARMA+PNN), qualitative model (BP-PNN), and random strategy which decide to buy or sell randomly. The hybrid model showed the best prediction accuracy with 75.3%. The agent simulation strategies also indicated the superiority of the hybrid model whose profitability exceeded 16.7%.
Xu and Wang (2024)	Solar PV Energy	Forecasting Related Study: Neural Networks, Decision Trees, SVM, K-means, Random Forests, and Gradient Boosting	Regarding m-Si silicon technologies, Model 1 prioritizes employee well-being. It shows strong predictive abilities (0.923 R^2 in training, 0.920 R^2 in testing) and low RMSE values (1.956 in training, 1.998 in testing). The Model 2 for p-Si silicon technologies focuses on supply chain ethics. The R^2 values of 0.934 in training and 0.933 in testing for this model. Finally, the a-Si silicon technologies utilize the bagging approach (Model 3) to assess its environmental impact. This approach yields excellent statistics, with high R^2 values of 0.930 in training and 0.929 in testing, as well as low RMSE values of 1.865 in training and 1.890 in testing.

Table 5 (continued)

Authors	Dependent Variable	Model Used in Study	Summary of the Findings
Xu et al. (2024)	Stock Market Index	Proposed New Model: Markow Switching based Heterogeneous Auto-Regressive	This method tested with emerging Chinese Stock Market (Shanghai Stock Exchange Market). The sample data included open price, close price, high price and low price, trading volumes and the value of transactions of the index. The effects of the CS liquidity on realized volatility were much stronger in crisis periods than non-crisis periods.
Yamaçli and Tunçsiper (2024)	Consumer Price Index	Forecasting Related Study: Johansen Cointegration Error Correction Model (ECM) Artificial Neural Network (ANN)	Johensen cointegration analysis indicated the statistically significant and negative relationship between unemployment rate and inflation. The positive relationship between oil prices and inflation is found with this method. ECM did not find the statistically significant relationship between the consumer prices and the unemployment rate in the short run. ANN showed better performances than the Johensen cointegration with 0.07 MAE, 0.09 RMSE, 0.01 MAPE, and 0.95 R^2.
Yilmaz and Yildiztepe (2024)	Stock Return	Forecasting Related Study: Temporal Convolution Networks (TCNs), RNNs, LTSM, and GRU models	The TCN model had the best out of sample forecasting performance with 93% accuracy at least, for all forecasting horizons of three stock returns. When the forecasting horizon was increased, the prediction performance of other models also increased.
Zhang and Zhang (2023)	Equity Premium	Proposed New Model: Averaging Method based on Forward-Validation	The performance of FVMA method is compared Mallows model averaging (MMA), JMA, LsoMA, multivariate leave-h-out cross-validation averaging (MCVAh), model with all the predictors (Full Model), lasso and random forest (RF). When all the candidate models are misspecified, the FVMA method combining 13 candidate models outperforms the other model averaging methods. It also outperforms Lasso and RF methods under every setting.
Zhang et al. (2024)	Crude Oil Prices	Forecasting Related Study: Dynamic Conditional Correlation-Multivariate Generalized Autoregressive Conditional Heteroskedasticity (DCC-MGARCH) Spatiotemporal Information Recombination Hypergraph Neural Network (STIR-HGNN)	The DCC-MGARCH model did not capture the intricate interdependence between energy and agricultural futures prices. The MAE statistics are 8.51%, 2.33%, and 1.42% for soybean, coffee, and cocoa prices, respectively. This statistics indicated that incorporating energy futures prices enhanced the forecasting performance of the STIR-HGNN model. The RMSE indicators were also in line with the MAE indicators. With the inclusion of energy prices to the model, the accuracy of the model increased 1.63%, 1.42%, and 0.68% for soybean, coffee, and cocoa prices, respectively.

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Declarations

Conflict of interest No conflicts to declare.

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